GGG–G–61

AUGUST 23, 1932

FEDERAL STANDARD STOCK CATALOGUE

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

GAGES; PLUG AND RING, PLAIN AND THREAD

This specification was approved for promulgation by the Federal Specifications Board on August 23, 1932, for the use of the departments and independent establishments of the Government in the purchase of this commodity, and shall become mandatory for all departments and independent establishments of the Government not later than May 15, 1933. It may be put into effect, however, at any earlier date, after promulgation.

A. APPLICABLE FEDERAL SPECIFICATIONS.

A–1. There are no other Federal specifications applicable to this specification.

B. TYPES, CLASSES, ETC.

B–1. Gages, in accordance with this specification, shall be furnished in the following types and classes as required:

Type A. Plain gages—
- Class (1). Plain cylindrical plug gages.
- Class (2). Plain ring gages.

Type B. Screw-thread gages—
- Class (1). Thread plug gages.
- Class (2). Thread ring gages.

C. MATERIAL AND WORKMANSHIP.

C–1. Gages of satisfactory quality shall be furnished. Manufacturers shall be given a sufficient range in the selection of materials to meet this requirement.

C–2. The workmanship shall be consistent with the tolerances hereinafter specified.

D. GENERAL REQUIREMENTS.

D–1. Gages shall be free from seams, scratches, cracks, rust spots, and other imperfections which may be detrimental to their use or durability.

D–2. The gaging surface of plain plug and ring gages shall show a minimum hardness of 63 on the Rockwell C scale.

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Thread plug and ring gages shall show a minimum hardness of 60 on the Rockwell C scale and shall show an equivalent hardness by file test on the gaging surfaces.

D–3. *Marking of gages.*—Each gage shall be plainly and permanently marked, for identification, with the size and the manufacturer's name or trade-mark. The size designation of screw-thread gages shall be in accordance with the practice established by the National Screw Thread Commission as follows:

The method of designating a screw thread by means of symbols shall be by the use of the initial letters of the thread series, preceded by the diameter in inches (or the screw number) and number of threads per inch, all in Arabic characters, and followed by the classification of fit in Arabic numerals. If the thread is left hand, the symbol "L. H." shall follow the number of threads. No symbol shall be used to distinguish right-hand threads. For screw threads of American National form but of special diameters, pitches, and lengths of engagement, the symbol "NS" shall be used. Examples:

	Mark
American National coarse-thread series: To specify a threaded part 1 in. diameter, 8 threads per inch, class 1 fit	1"–8–NC–1
American National fine-thread series: Threaded part 1 in. diameter, 14 threads per inch, class 3 fit	1"–14–NF–3
American National form, special pitch: Threaded part 1 in. diameter, 12 threads per inch, class 2 fit	1"–12–NS–2
American National form, left-hand thread: Threaded part 1 in. diameter, 12 threads per inch, class 3 fit	1"–12 L. H.–NS–3

The number of threads per inch shall be indicated in all cases, irrespective of whether or not it is the standard number of threads for that particular size of threaded part.

D–4. *Standard temperature.*—The nominal dimensions of gages shall be within the specified limits at a temperature of 68° F. (20° C.).

E. DETAIL REQUIREMENTS.

E–1. *Type A, plain gages.*

E–1a. *Type A, class (1)—Plain cylindrical plug gages.*—Plain cylindrical plug gages shall be of the taper lock design for the range from 0.059 to and including 1.510 inches, and of the reversible design with reversible gaging members for the range from above 1.510 to and including 4.510 inches, as shown in Figure 1, except that the use of the taper lock design is optional to the purchaser for the size range from above 1.510 to and including 2.510 inches.

E–1a(1). *Details of construction, plug gage blanks of taper lock design, above 0.059 to and including 1.510 inches.*—Details of design and dimensions of taper lock plug gage blanks shall be in accordance with Tables I and II.

E–1a(2). *Details of construction, plug gage blanks of reversible design, above 1.510 to and including 4.510 inches.*—Details of design and dimensions of reversible plug gages shall be in accordance with Tables III and IV.

E–1a(3). *Details of construction, handles for plain cylindrical plug gages.*—Handles for both taper lock and reversible plain cylindrical plug gages shall be of the hexagonal type in accordance with Tables V and VI, and Figure 2.

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E–1a(4). *Wear allowances and tolerances on gaging diameter, plain cylindrical plug gages.*—Gages shall be made with wear allowances and within gage makers' tolerances in accordance with Table VII of this specification.

E–1b. *Type A, class (2), plain ring gages.*—As shown in Figure 3, in plain ring gages above 0.059 to and including 0.510 inch, a hardened bushing pressed into a soft gage body shall be used in place of the one-piece ring gage. The one-piece gage shall be employed in all cases above 0.510 inch. Gages in sizes above 1.510 inches shall be flanged in order to eliminate unnecessary weight and facilitate handling.

An annular groove shall be provided in the periphery of "not go" blanks as a means of identification.

E–1b(1). *Details of construction, plain ring gages.*—Details of design, and dimensions of plain ring gages shall be in accordance with Tables VIII and IX.

E–1b(2). *Wear allowances and tolerances on gaging diameter, plain ring gages.*—Plain ring gages shall be made with wear allowances and within gage makers' tolerances in accordance with Table VII in this specification.

E–2. *Type B, screw thread gages.*

E–2a. *Type B, class (1), thread plug gages.*—Thread plug gages shall be of the taper lock design for the range from 0.059 to and including 1.510 inches and of the reversible design with reversible gaging members for the range from above 1.510 to and including 4.510 inches. General details of construction are shown in Figure 4.

E–2a(1). *Details of construction, thread plug gage blanks of taper lock design, above 0.059 to and including 2.510 inches.*—Details of design and dimensions of taper lock thread plug gage blanks shall be in accordance with Tables X and XI.

E–2a(2). *Details of construction, thread plug gage blanks of reversible design, above 1.510 to and including 4.510 inches.*—Details of design and dimensions of reversible thread plug gage blanks shall be in accordance with Tables XII and XIII.

E–2a(3). *Details of construction, handles for thread plug gages.*—Handles for both taper lock and reversible thread plug gages shall be of the hexagonal type, in accordance with Tables XIV and XV and Figure 5.

E–2a(4). *Gaging dimensions and tolerances, thread plug gages.*—Thread-plug gages shall be class X, in accordance with tolerances as specified in Table XVI, unless otherwise specified. See note I–2.

The gaging dimensions shall be as follows:

E–2a(4)a. *Angle of thread.*—The included angle of thread shall be 60° unless otherwise specified.

E–2a(4)b. *Pitch.*—The pitch shall conform to the threads per inch specified in the order.

E–2a(4)c. *Major diameter.*—The major diameter of "go" gages shall correspond to a width of flat at the crest of one-eighth of the pitch. The major diameter of "not go" gages shall be at least 0.0002 inch less than that specified for "go" plug gages.

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E–2a(4)d. *Pitch diameter.*—The pitch diameter of "go" gages shall be the basic pitch diameter of the thread to be gaged. The pitch diameter of "not go" gages shall be the maximum pitch diameter of the thread to be gaged.

E–2a(4)e. *Minor diameter.*—A relief shall be provided at the root of the "go" plug gage, the width of which shall be not greater than one-eighth of the pitch. A relief shall be provided at the root of the "not go" plug gage, the width of which shall be approximately one-fourth of the pitch. See Figure 6.

E–2b. *Type B, class (2), thread ring gages.*

E–2b(1). *Details of construction, thread ring gages.*—Thread ring gages shall be adjustable, of the types and with the locking device of the single-unit type shown in Figure 7, as follows:

A thin flat disk type with one adjusting slot for all sizes and pitches, both "go" and "not go," No. 0 to 5/16 inch, inclusive.

A thin flat disk type with two adjusting slots for the following: (*a*) All sizes and pitches, "go" and "not go," above 5/16 to and including ½ inch; (*b*) fine pitches,[1] "go" and "not go," above ½ to and including 4½ inches; (*c*) coarse pitches, "not go" only, above ½ to and including 4½ inches.

A thick flanged type with two adjusting slots for all "go" coarse pitch gages, above ½ to and including 4½ inches.

The dimensions of thread ring gage blanks in the range from No. 0 to 4½ inches, inclusive, and of parts for the thread ring gage locking device, shall be in accordance with Tables XVII, XVIII, XIX, and XX.

E–2b(2). *Gaging dimensions and tolerances, thread ring gages.*—Thread ring gages shall be class X in accordance with tolerances as specified in Table XVI unless otherwise ordered. See note I–2.

Thread setting plugs shall be used to govern the size of thread ring gages. The gaging dimensions of thread ring gages and setting plugs shall be as follows:

E–2b(2)a. *Angle of thread.*—The included angle of thread shall be 60° unless otherwise specified.

E–2b(2)b. *Pitch.*—The pitch shall conform to the threads per inch specified in the order.

E–2b(2)c. *Major diameter.*—A relief shall be provided at the root of the "go" ring gage, the width of which shall be not greater than one-eighth of the pitch. A relief shall be provided at the root of the "not go" ring gage, the width of which shall be approximately one-fourth of the pitch. See Figure 6.

The maximum major diameter of the thread setting plug for the "go" ring gage shall be the maximum major diameter of the screw to be gaged. The minimum major diameter of the thread setting plug for the "not go" gage shall be the minimum major diameter of the screw to be gaged.

E–2b(2)d. *Pitch diameter.*—The pitch diameters of "go" thread setting plugs or ring gages shall be the maximum pitch diameter of

[1] Specific information as to the meaning of the terms "fine pitches" and "coarse pitches" as used above is given in the footnote to Table XVII.

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the screw to be gaged. The pitch diameters of "not go" thread setting plugs or ring gages shall be the minimum pitch diameter of the screw to be gaged.

E–2b(2)e. *Minor diameter.*—The maximum minor diameter of the "go" ring gage shall be the minimum diameter specified for the tapped hole to be gaged. The minimum minor diameter of the "not go" ring gage shall be at least 0.0002 inch greater than that specified for "go" ring gages.

F. METHODS OF INSPECTION.

F–1. *Finish of gaging surfaces.*—Gaging surfaces shall be inspected visually as to seams, cracks, scratches, rust spots, and other defects which may materially affect the life of the gage.

F–2. *Gaging dimensions.*—The gaging dimensions of all gages for which tolerances are specified herein shall be measured. The accuracy of such measurements shall be controlled by comparison with dimensional standards which have been certified by the Bureau of Standards.

F–2a. For measurement of the pitch diameter of thread plug gages, wire methods shall be used. (Such methods are described in Appendix 2 of the 1928 Report of the National Screw Thread Commission, Bureau of Standards Miscellaneous Publication No. 89.) In measuring the pitch diameter of screw thread gages by means of wires, and in measuring the wires themselves, the same contact pressure shall be used, and wires shall be measured between a flat contact and a cylindrical contact 0.750 inch in diameter. Unless otherwise specified a contact pressure of 1 pound shall be used for all wires and gages for pitches 80 threads per inch and coarser.

F–2b. The pitch diameter of thread ring gages—that is, the adjustment of size—shall be determined by the fit of the ring gage on a thread setting plug.

F–3. *Taper lock handles and gaging members.*—The taper shanks and handles of gages of taper lock design shall be inspected by means of gages made in accordance with Tables XXI and XXII.

G. PACKAGING, PACKING, AND MARKING.

G–1. *Packaging.*—In the shipment of gages, extreme precautions shall be taken against both corrosion and damage by contact with other gages during transit. All defining steel surfaces shall be greased and protected with waxed paper. Newspaper, wrapping paper (unwaxed) or excelsior shall not be allowed to come into contact with gaging surfaces. Small gages suitably wrapped may be separated by newspapers, excelsior, etc. Heavy gages shall be rigidly fastened in place in a strong, rigid container so that no movement is possible. Plain plug and ring gages shall not be shipped mated. When large thread-ring gages are shipped with their master or setting thread-plug gages, mating is desirable as a means of protecting the plug threads against damage. In such cases a non-acid grease shall be used, such that electrolytic corrosion between the mating gage surfaces will be prevented.

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G-2. *Packing.*—Unless otherwise specified, gages shall be delivered in standard commercial containers so constructed as to insure acceptance by common or other carriers for safe transportation at the lowest rate to the point of delivery.

G-3. *Marking.*—Unless otherwise specified, shipping containers shall be marked with the name, size, and quantity of gages contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS.

H-1. The following departmental specifications of the issue in effect on date of invitation for bids shall form a part of this specification:

H-1a. Navy Department, General Specifications for Inspection of Material, copies of which may be obtained without cost on application to Bureau of Supplies and Accounts, Navy Department, Washington, D. C.

I. NOTES.

I-1. *Terminology.*—The following definitions are intended to clarify the meaning of certain technical terms employed in this specification:

A *plain cylindrical plug gage* is a complete unthreaded internal gage of single or double ended type for the size control of holes. It consists of handle and gaging member or members, with suitable locking means.

A *progressive cylindrical plug gage* is a complete unthreaded internal gage consisting of handle and gaging member in which the "go" and "not go" gaging sections are combined in a single unit secured to one end of the handle.

A *thread plug gage* is a complete internal thread gage of either single or double ended type, comprising handle and threaded gaging member or members, with suitable locking means.

The *gaging member* is that integral unit of a plug gage which is accurately finished to size and is employed for size control of the work. In taper lock gages, the gaging member consists of a shank and a gaging section.

The *gaging section* is that portion of the gage which comes into physical contact with the work. In the plug gage range above 1.510 to and including 4.510 inches, the gaging section is identical with the gaging member.

The *shank* (applied to taper lock gages only) is that portion of the gaging member which is employed for fixing the gaging member to the handle.

The term *"taper lock"* designates that construction in which the gaging member has a taper shank, which is forced into a taper hole in the handle. This design is standard for plug gages in the range above 0.059 inch to and including 1.510 inches, and is optional for plain cylindrical and thread plug gages in the range above 1.510 inches to and including 2.510 inches.

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A *reversible plug gage* is a plain cylindrical or thread plug gage, in which three wedge-shaped *locking prongs* on the handle are forced into corresponding *locking grooves* in the gaging member by means of a single through screw thus providing a self-centering support with a positive lock. This design is standard for all plug gages in the ranges above 1.510 to and including 4.510 inches.

Lightening holes are unfinished drilled holes provided in the heavier sizes of gaging members for the sole purpose of reducing the weight of the gage.

The *handle* is that portion of a plug gage which is employed as supporting means for the gaging member or members.

The *drift hole* or *drift slot* is a small hole or slot provided in the side of a taper lock gage handle near the "go" end through which a pin or drift may be inserted for the purpose of ejecting the gaging member from the handle.

A *plain ring gage* is an unthreaded external gage employed for the size control of external diameters. In the smaller sizes it consists of a gage body into which is pressed a *bushing*, the latter being accurately finished to size for gaging purposes.

A *thread ring gage* is an external thread gage employed for the size control of threaded work, means of adjustment being provided integral with the gage body.

The *flange* is that external portion of a large ring gage which is reduced in section for the purpose of lightening the gage.

The *hub* is the mid-section of a flanged ring gage. It determines the length of the gaging section.

The *thread ring gage locking device* is a means of expanding and contracting the thread ring gage during the manufacturing or resizing processes. It also provides an effectual lock. It comprises an *adjusting screw*, a *locking screw*, and a *sleeve*. For detailed description and illustration see page 34.

Adjusting slots are radial slots provided in thread ring gages in order to facilitate expansion and contraction of gage size by means of the adjusting device. An adjusting slot always terminates in an *adjusting slot terminal hole*.

The *locking slot* is that slot which passes entirely through the wall of a thread ring gage. In conjunction with the thread ring gage locking device, it permits expansion and contraction of gage size.

I-2. *Thread gage tolerances.*—When a wear allowance on "go" thread gages which is not provided by class X gages is desired by the purchaser, such a suitable allowance may be specified. For this purpose the National Screw Thread Commission has established classes Y and Z gages for certain classes of fit, the tolerances for which are given in Tables XXIII and XXIV in this specification.

I-3. *Samples.*—It is believed that this specification is adequate and that normally no samples will be necessary to determine compliance with the requirements, prior to award. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

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I–4. *Precaution.*—In preparing bid requests or orders special attention should be given to furnishing adequate information as to gaging dimensions, particularly for thread gages as indicated under E–2a(4), E–2b(2), etc.

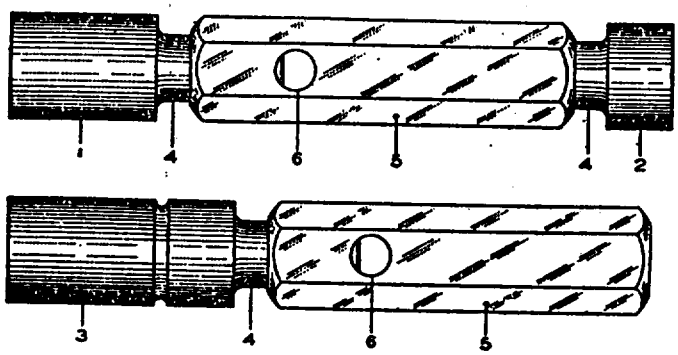
I–5. This specification governs all United States Government purchases of this commodity. Users are requested to retain this specification in permanent file until a revision is issued.

I–6. Interested parties are advised that an Alphabetical Index of Federal Specifications may be purchased as noted in the paragraph next below, price to be obtained from the Superintendent of Documents.

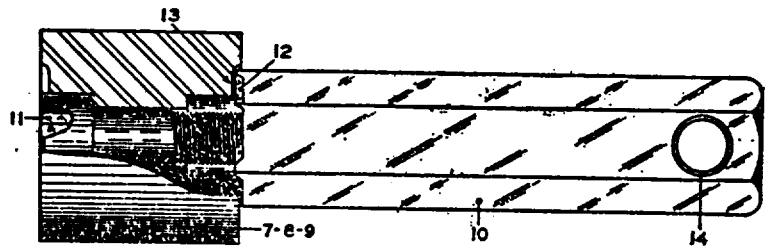
I–7. Copies of this specification may be obtained upon application, accompanied by money order, or coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington, D. C. Price 10 cents.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

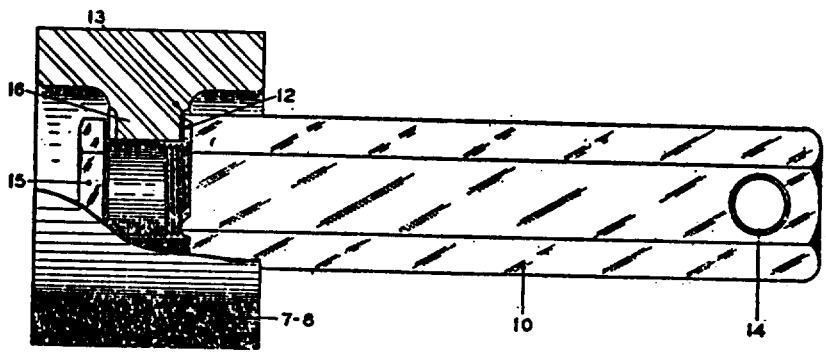
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Range: Above 0.059 to and including 1.510 inches; optional above 1.510 to and including 2.510 inches



Range: Above 1.510 to and including 2.510 inches



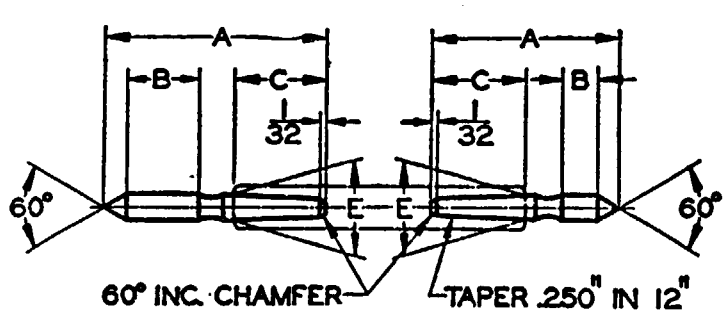
Range: Above 2.510 to and including 4.510 inches

FIGURE 1.—American gage design standard plain cylindrical plug gages, details of construction

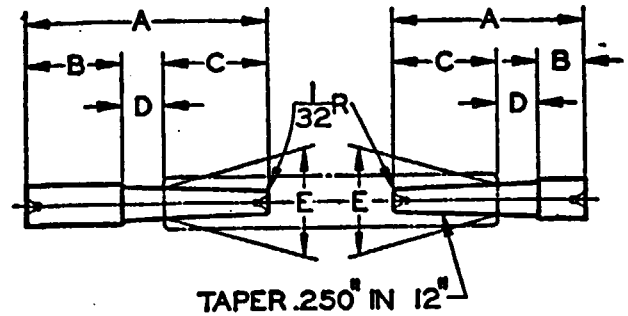
1. "Go" gaging member.
2. "Not go" gaging member.
3. Progressive gaging member.
4. Shank.
5. Taper lock handle.
6. Drift hole (or slot).
7. "Go" gaging member.
8. "Not go" gaging member.
9. Progressive gaging member.
10. Handle for reversible gage.
11. Socket head screw.
12. Locking prong.
13. Locking groove.
14. Cross-pin hole.
15. Hexagon head screw.
16. Web.

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TABLE I.—*Plain cylindrical plug gaging members, taper lock design, range above 0.059 to and including 0.240 inch*



ABOVE 0.059 TO AND INCLUDING 0.150 INCH

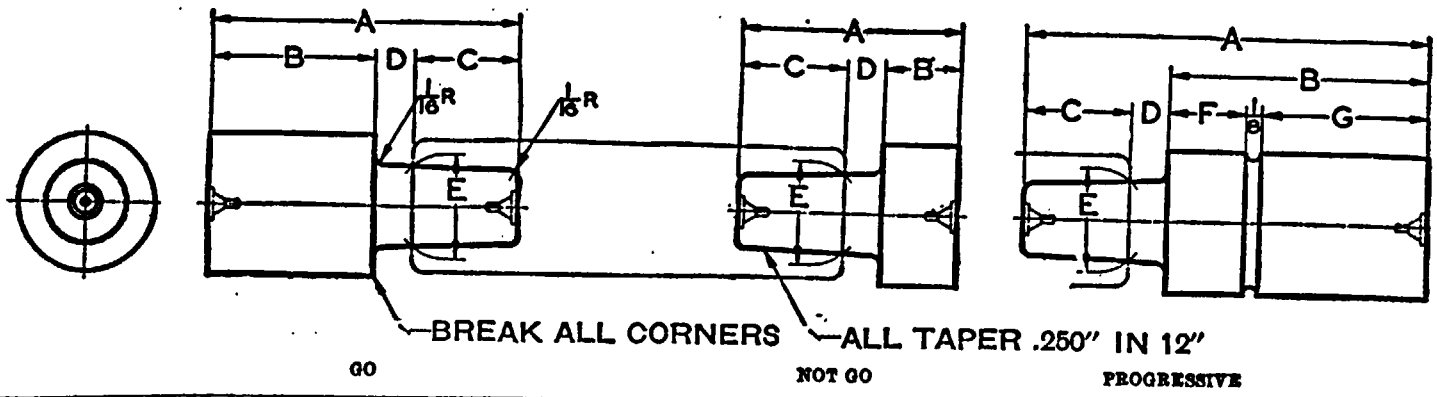


ABOVE 0.150 TO AND INCLUDING 0.240 INCH

Handle size No.	Range in diameters		General dimensions											
			Go						Not go					
	Above—	To and including—	A	B	C	D	E		A	B	C	D	E	
							Min.	Max.					Min.	Max.
	Inch	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*
000	0.059	0.105	1 5/32	3/8	1/2	---	0.125	0.126	3 1/32	3/16	1/2	---	0.125	0.126
00	.105	.150	1 11/32	7/16	9/16	---	.155	.156	1 1/4	3/32	9/16	---	.155	.156
0	.150	.240	1 15/32	1 9/32	5/8	1/4	.180	.181	1 5/32	9/32	5/8	1/4	.180	.181

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TABLE II.—*Plain cylindrical plug gaging members, taper lock design, range above 0.240 to and including 1.510 inches*

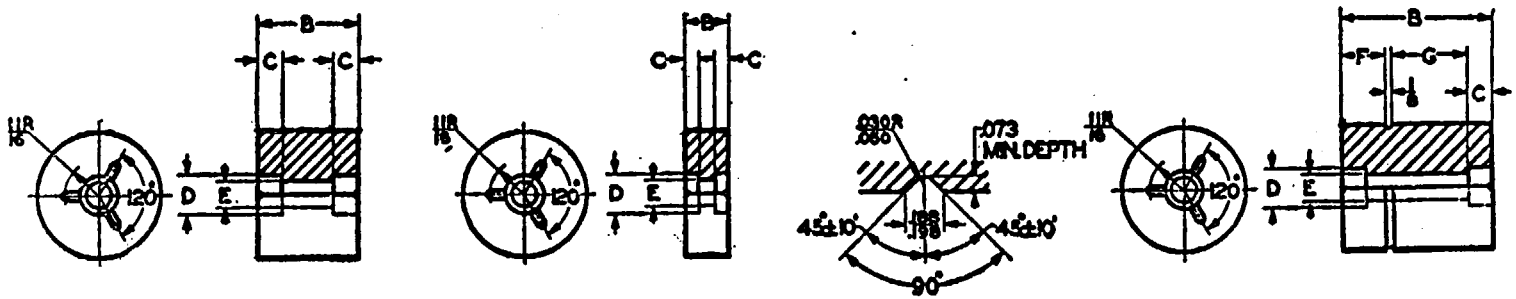


Handle size No.	Range in diameters		General dimensions																			
			Go						Not go						Progressive							
	Above—	To and including—	A	B	C	D	E		A	B	C	D	E		A	B	C	D	E		F	G
							Min.	Max.					Min.	Max.					Min.	Max.		
	Ins.	Ins.	Ins.	Ins.	In.	In.	In.	In.	Ins.	In.	In.	In.	In.	In.	Ins.	Ins.	In.	In.	In.	In.	In.	Ins.
1	0.240	0.365	1¾	⅜	⅝	¼	0.239	0.240	1 5/16	5/16	⅜	¼	0.239	0.240	2 5/16	1 5/16	⅝	¼	0.239	0.240	5/16	¾
2	.365	.510	2	1	⅝	¼	.309	.310	1⅜	⅜	⅝	¼	.309	.310	2½	1½	⅝	¼	.309	.310	⅜	1
3	.510	.825	2¼	1¼	¾	¼	.408	.410	1½	½	¾	¼	.408	.410	2⅞	1⅞	¾	¼	.408	.410	½	1¼
4	.825	1.135	2 9/16	1⅜	⅞	5/16	.608	.610	1 13/16	⅝	⅞	5/16	.608	.610	3 5/16	2⅛	⅞	5/16	.608	.610	⅝	1⅝
5	1.135	1.510	2⅞	1½	1	⅜	.808	.810	2⅛	¾	1	⅜	.808	.810	3¾	2⅜	1	⅜	.808	.810	¾	1½
5, optional, see note	1.510	2.010	3¼	1⅞	1	⅜	.808	.810	2¼	⅞	1	⅜	.808	.810	4¼	2⅞	1	⅜	.808	.810	⅞	1⅞
5, optional, see note	2.010	2.510	3⅜	2	1	⅜	.808	.810	2¼	⅞	1	⅜	.808	.810	4⅜	3	1	⅜	.808	.810	⅞	2

NOTE.—The use of taper lock plain cylindrical plug gaging members and handles is optional in the range above 1.510 to and including 2.510 inches, but the use of the reversible design is standard for all sizes above 1.510 inches.

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TABLE III.—*Plain cylindrical plug gaging members, reversible design, range above 1.510 to and including 2.510 inches*

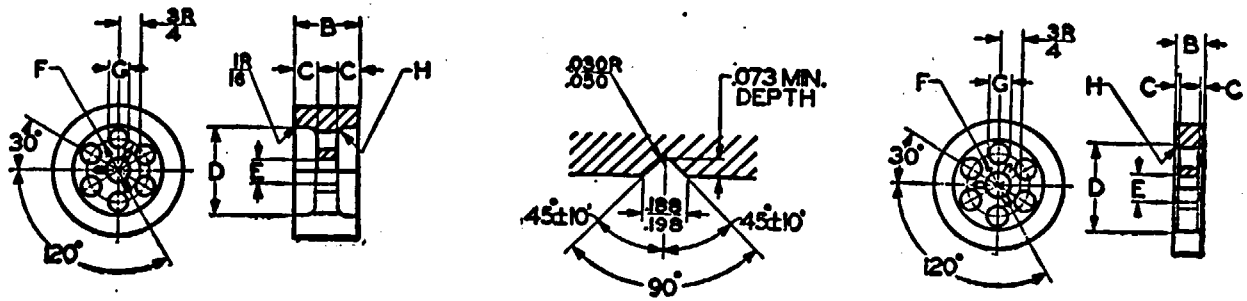


ENLARGED SECTION OF
LOCKING GROOVE
8 REQUIRED ON BOTH ENDS
OF ALL GAGES

Handle size No.	Plain plug diameters				Go				Not go				Progressive					
	Nominal range, inclusive		Decimal range		*B*	*C*	*D*	*E*	*B*	*C*	*D*	*E*	*B*	*C*	*D*	*E*	*F*	*G*
	From—	To—	Above—	To and including—														
	Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*
6	1½	2	1.510	2.010	1⅞	½	25/32	17/32	⅞	9/32	25/32	17/32	2⅞	½	25/32	17/32	⅞	1⅛
6	2	2½	2.010	2.510	2	½	25/32	17/32	⅞	9/32	25/32	17/32	3	½	25/32	17/32	⅞	2

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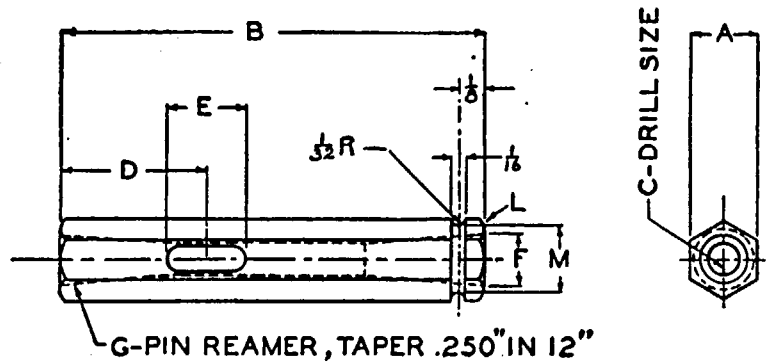
TABLE IV.—*Plain cylindrical plug gaging members, reversible design, range above 2.510 to and including 4.510 inches*



Handle size No.	Plain plug diameters				Go							Not go						
	Nominal range, inclusive		Decimal range		B	C	D	E	F	G	H	B	C	D	E	F	G	H
	From—	To—	Above—	To and including—														
	Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inches*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inch*	*Inches*	*Inch*	*Inch*
7	2½	3	2.510	3.010	2⅛	1 1/16	1⅞	2 9/32	------	------	5/16	1	1/8	1⅞	2 9/32	------	------	5/16
7	3	3½	3.010	3.510	2¼	¾	2¼	2 9/32	------	------	5/16	1	1/8	2¼	2 9/32	------	------	5/16
7	3½	4	3.510	4.010	2¼	¾	2⅝	2 9/32	------	------	5/16	1	1/8	2⅝	2 9/32	------	------	5/16
7	4	4½	4.010	4.510	2¼	¾	3	2 9/32	1 1/16	¾	5/16	1	1/8	3	2 9/32	1 1/16	¾	5/16

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TABLE V.—*Plain cylindrical plug gage handles, taper lock design, range above 0.059 to and including 0.240 inch*

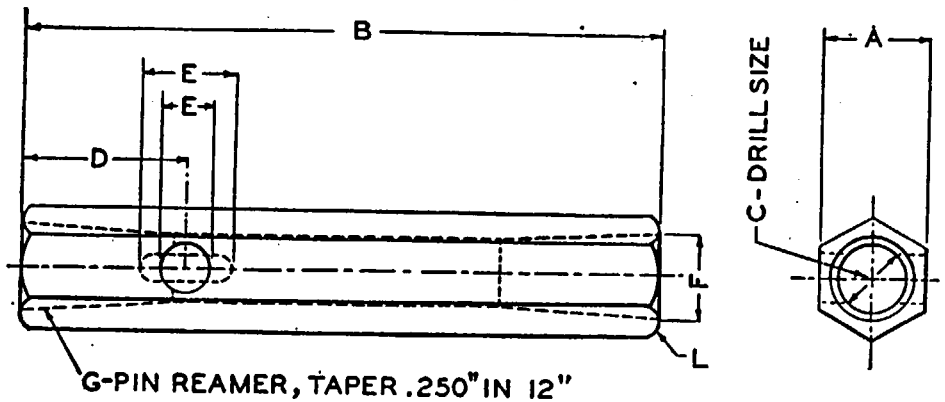


Handle size No.	Range, plain plug diameters		General dimensions										
	Above—	To and including—	*A*	*B*	*C* Drill size	*D*	*E*	*F*		*G*	*L*	*M*	
								Min.	Max.			Min.	Max.
	Inch	*Inch*	*Inch*	*Inches*		*Inch*	*Inch*	*Inch*	*Inch*	*No.*	*Inch*	*Inch*	*Inch*
000	0.059	0.105	3/16	1 1/2	No. 34 (0.111)	9/16	9/64 by 1/4	0.125	0.126	000	1/32	0.172	0.177
00	.105	.150	1/4	1 3/4	No. 29 (0.136)	5/8	3/32 by 5/16	.155	.156	0	1/32	.235	.240
0	.150	.240	5/16	2	No. 20 (0.161)	1 1/16	1/8 by 3/8	.180	.181	2	1/32	.297	.302

NOTE.—The purpose of the groove in the "not go" end of the handle is to distinguish the "not go" from the "go" end.

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TABLE VI.—*Plain cylindrical plug gage handles taper lock design, range above 0.240 to and including 1.510 inches*



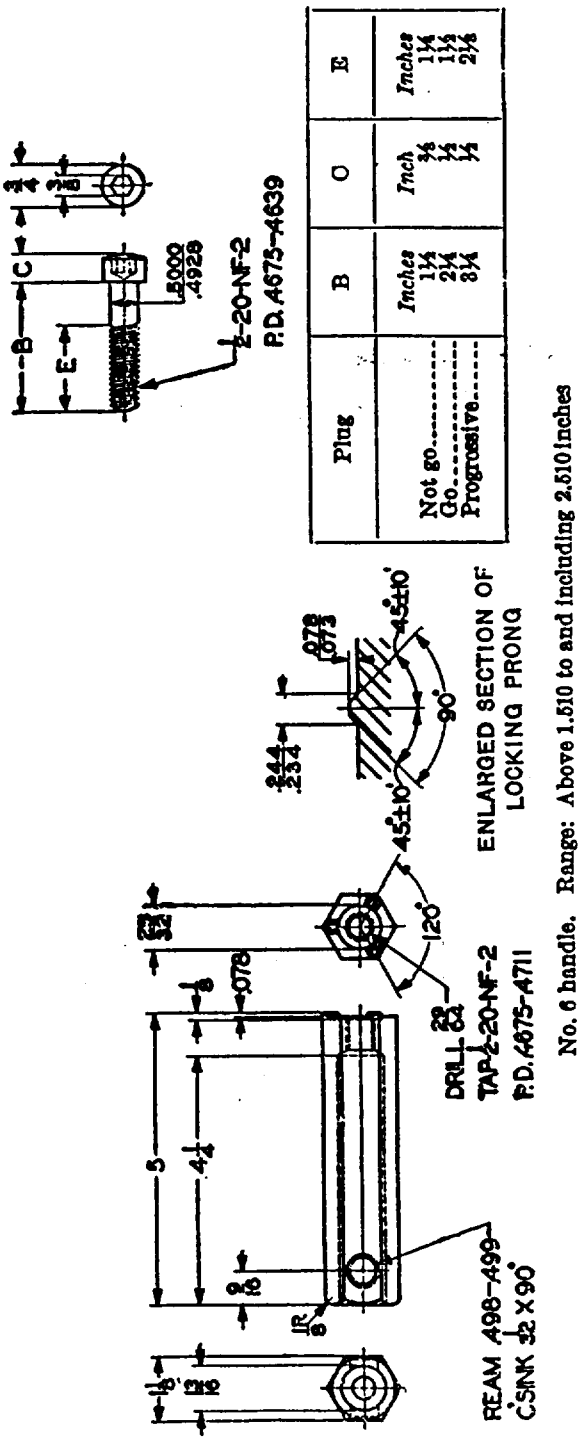
Size	Range, plain plug diameters		General dimensions								
	Above—	To and including—	*A*	*B*	*C* Drill size	*D*	*E*	*F* Min.	*F* Max.	*G*	*L*
	Inches	*Inches*	*Inches*	*Inches*		*Inches*	*Inch*	*Inch*	*Inch*	*No.*	*Inch*
1	0.240	0.365	3/8	2 3/4	7/32	2 5/32	1/8 by 1/2	0.239	0.240	4	1/16
							Diameter				
2	.365	.510	1/2	3	L (0.290)	2 5/32	15/64	.309	.310	6	1/16
3	.510	.825	11/16	3 1/4	25/64	27/32	13/32	.409	.410	7	3/32
4	.825	1.135	7/8	3 5/8	27/64	6 3/64	3/8	.609	.610	10	3/32
5	1.135	1.510	1 1/8	4	29/32	1 1/8	7/16	.809	.810	11	1/8
5, optional, see note	1.510	2.510	1 1/8	4	29/32	1 1/8	7/16	.809	.810	11	1/8

NOTE.—The use of taper lock plain cylindrical plug gaging members and handles is optional in the range above 1.510 to and including 2.510 inches, but the use of the reversible design is standard for all sizes above 1.510 inches.

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FEDERAL STANDARD STOCK CATALOGUE

(Section IV, Part 5)



Plug	B	C	E
	Inches	*Inch*	*Inches*
Not go	1¼	⅜	1¼
Go	2¼	½	1½
Progressive	3¼	½	2½

No. 6 handle. Range: Above 1.510 to and including 2.510 inches

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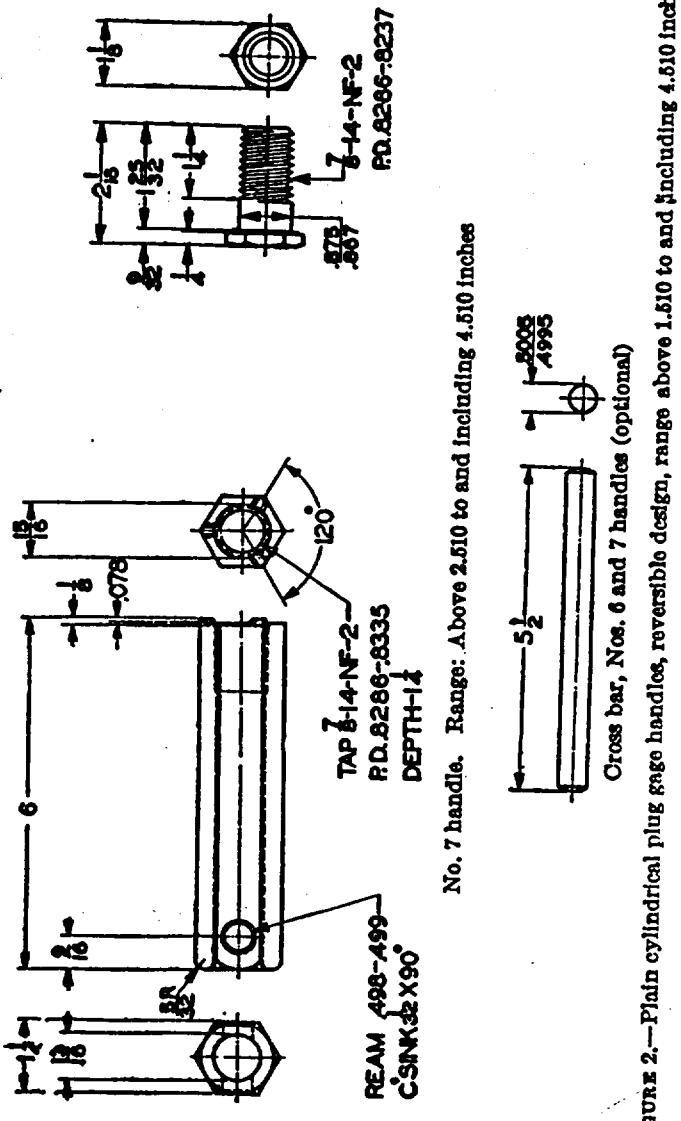
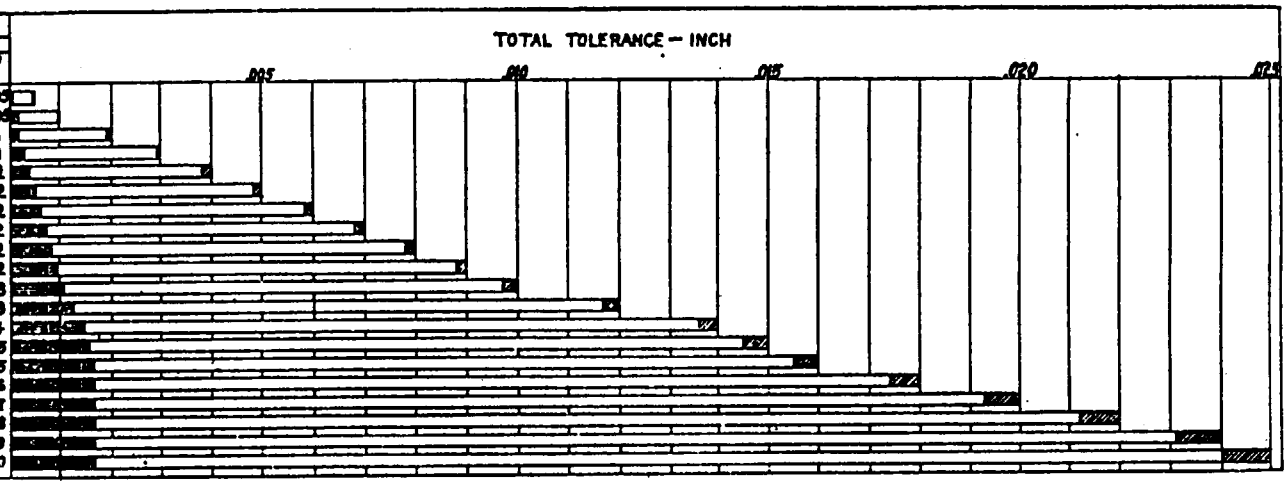


FIGURE 2.—Plain cylindrical plug gage handles, reversible design, range above 1.510 to and including 4.510 inches

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TABLE VII.—*Wear allowances and gagemakers' tolerances, plain plug and ring inspection gages, including plain ring gages for major diameter of screws, and plain plug gages for minor diameter of nuts*

COMPONENT TOTAL TOLERANCE	GAGE WEAR ALLOWANCE	GAGE TOLERANCES GO	GAGE TOLERANCES NOT GO
	1	2	3
.0005	.0000	.0001	.00005
.001	.0001	.0001	.00005
.002	.0001	.0001	.0001
.003	.0001	.0002	.0001
.004	.0002	.0002	.0002
.005	.0003	.0002	.0002
.006	.0004	.0002	.0002
.007	.0004	.0003	.0002
.008	.0005	.0003	.0002
.009	.0005	.0004	.0002
.010	.0005	.0005	.0003
.012	.0006	.0006	.0003
.014	.0006	.0008	.0004
.015	.0006	.0009	.0005
.016	.0006	.0010	.0005
.018	.0006	.0010	.0006
.020	.0006	.0010	.0007
.022	.0006	.0010	.0008
.024	.0006	.0010	.0009
.0252	.0006	.0010	.0010



NOTES

This chart shows, both numerically and graphically, wear allowances and gagemakers' tolerances for inspection gages. For component tolerances not listed, the next smaller tolerance shall be used.

The first four columns show numerical values, while the following portion of the table shows graphic values. Column 1 shows the total tolerance on the product, column 2 the wear allowance on the "go" gage, column 3 the tolerance on the "go" gage, and column 4 the tolerance on the "not go" gage. There is no wear allowance on the "not go" gage because it wears toward the "go" gage and in the direction of safety.

SYMBOLS

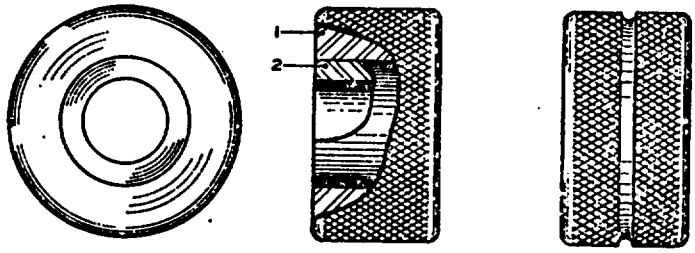
Represents wear allowance on the "go" gage.

Blank space represents remaining manufacturing tolerance on the product after the three factors—wear allowance and "go" and "not go" gage tolerances—have been deducted from the total manufacturing tolerance.

Represents gagemakers' tolerance on the "go" gage.

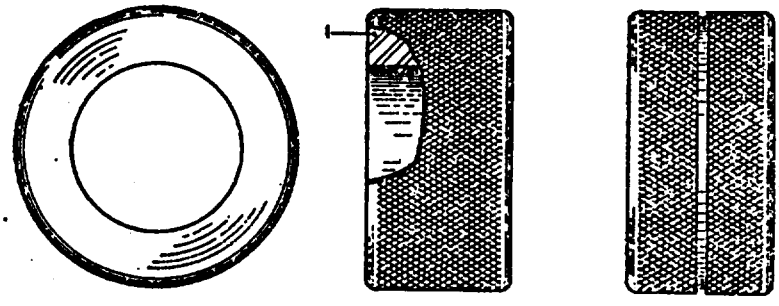
Represents gagemakers' tolerance on the "not go" gage.

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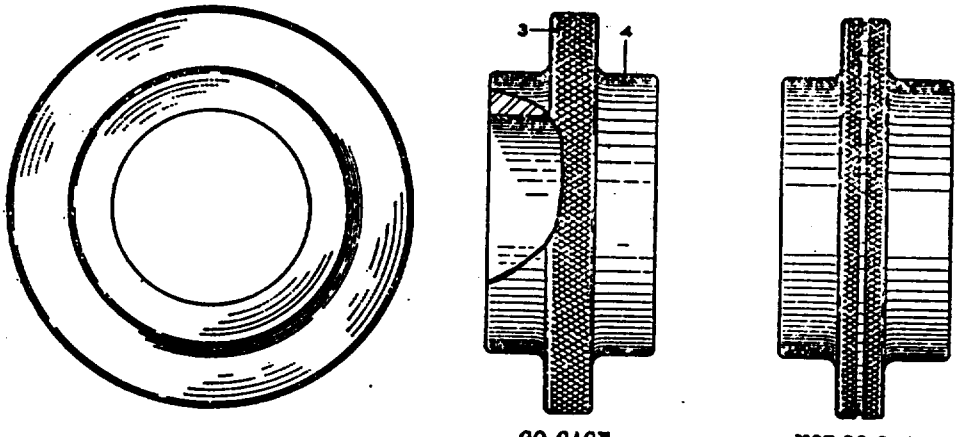
GO GAGE NOT GO GAGE

Range: Above 0.059 to and including 0.510 inch



GO GAGE NOT GO GAGE

Range: Above 0.510 to and including 1.510 inches



GO GAGE NOT GO GAGE

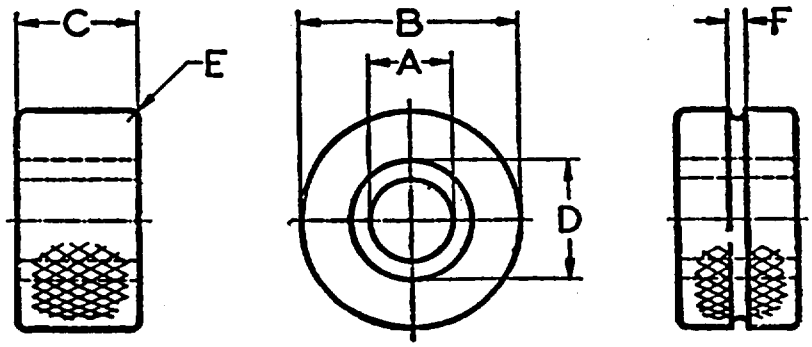
Range: Above 1.510 to and including 4.510 inches

1. Body.
2. Bushing.
3. Flange.
4. Hub.

FIGURE. 8.—American gage design standard plain ring gages, details of construction

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TABLE VIII.—*Plain ring gages, range above 0.059 to and including 1.510 inches*

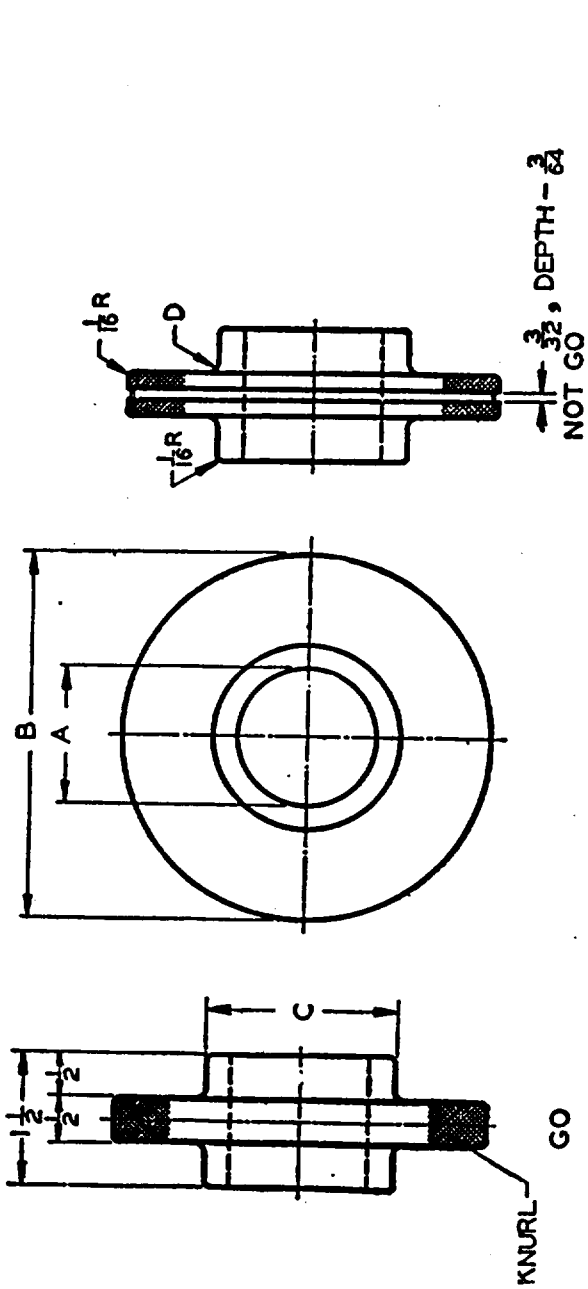


Ring size No.	A Range		General dimensions					
	Above—	To and including—	B Outside diameter	C Thickness	D Bushing diameter	E Radius	F "Not go" groove width	Length of bushing
	Inches	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*	*Inch*	
00	0.059	0.150	1 5/16	3/16	3/8	1/32	1/32	(1)
0	.150	.240	1 5/16	3/16	7/16	1/32	1/16	(1)
1	.240	.365	1 1/8	9/16	9/16	1/16	3/32	(1)
2	.365	.510	1 3/8	3/4	3/4	1/16	3/32	(1)
3	.510	.825	1 3/4	1 5/16	(2)	3/32	3/32	(2)
4	.825	1.135	2 1/8	1 1/8	(2)	3/32	3/32	(2)
5	1.135	1.510	2 1/2	1 5/16	(2)	3/32	3/32	(2)

[1] Bushings are 1/16 inch longer than ring thickness, but are ground flush after hole is finished.
[2] Sizes 3, 4, and 5 are solid.

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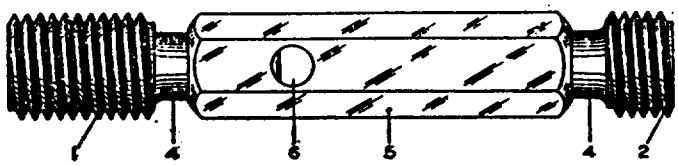
TABLE IX.—*Plain ring gages, range above 1.510 to and including 4.510 inches*



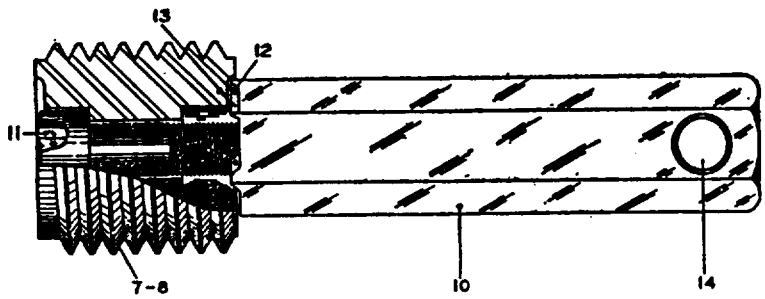
Ring size No.	A Range		B Outside diameter	C Hub diameter	D Radius
	Above—	To and including			
	Inches	Inches	Inches	Inch	Inch
6	1.510	2.010	4 1/16	A+ 7/16	1/16
7	2.010	2.510	4 5/16	A+ 7/16	1/16
8	2.510	3.010	5 1/2	A+1	5/16

Ring size No.	A Range		B Outside diameter	C Hub diameter	D Radius
	Above—	To and including			
	Inches	Inches	Inches	Inches	Inch
9	3.010	3.510	6 1/16	A+1	5/16
10	3.510	4.010	6 5/8	A+1 1/16	1/4
11	4.010	4.510	7 1/4	A+1 1/16	1/4

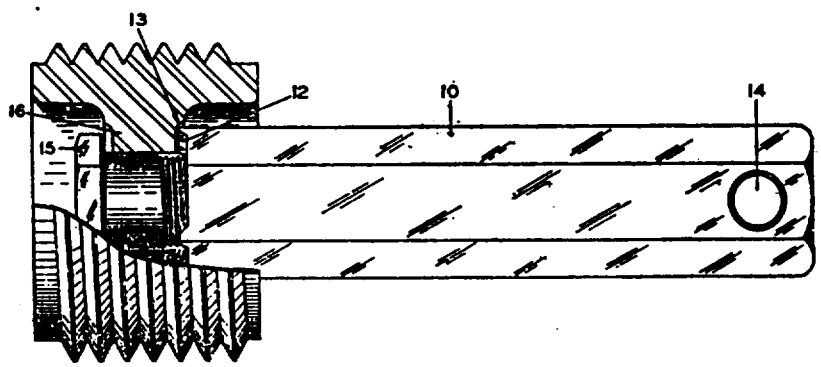
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Range: No 0 to and including 1.510 inches



Range: Above 1.510 to and including 2.510 inches



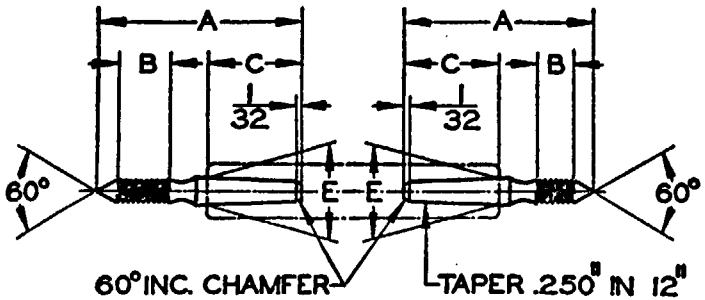
Range: Above 2.510 to and including 4.510 inches

FIGURE 4.—American gage design standard thread plug gages, details of construction

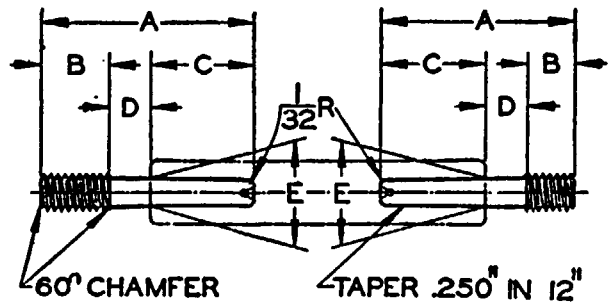
1. "Go" gaging member.
2. "Not go" gaging member.
4. Shank.
5. Taper lock handle.
6. Drift hole (or slot).
7. "Go" gaging member.
8. "Not go" gaging member.
10. Handle for reversible gage.
11. Socket head screw.
12. Locking prong.
13. Locking groove.
14. Cross-pin hole.
15. Hexagon head screw.
16. Web.

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TABLE X.—*Thread plug gaging members, taper lock design, range No. 0 to No. 12, inclusive*



NO. 0 TO NO. 6, INCLUSIVE

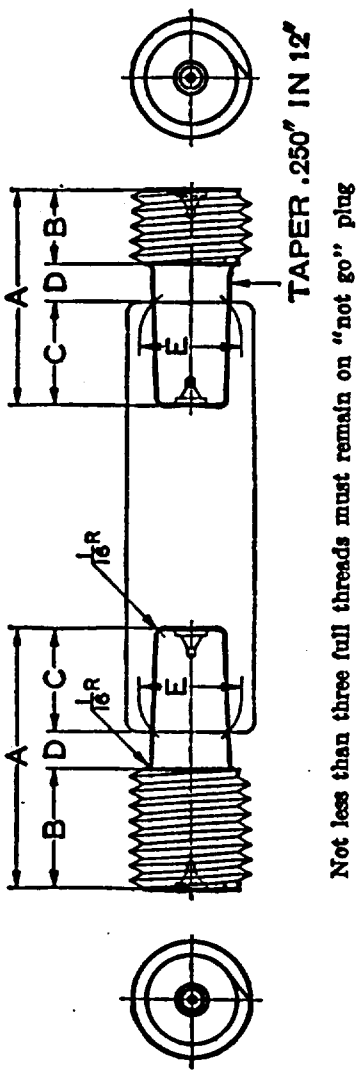


NO. 8 TO NO. 12, INCLUSIVE

Handle size No.	Range, thread plug diameters				General dimensions											
	Nominal, inclusive		Decimal		Go						Not go					
	From—	To—	Above—	To and including—	*A*	*B*	*C*	*D*	*E*		*A*	*B*	*C*	*D*	*E*	
									Min.	Max.					Min.	Max.
	No.	*No.*	*Inch*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*
000	0	3	0.059	0.105	1 1/32	1/4	1/2	-------	0.125	0.126	3 1/32	3/16	1/2	-------	0.125	0.126
00	4	6	.105	.150	1 7/32	5/16	9/16	-------	.155	.156	1 1/8	7/32	9/16	-------	.155	.156
0	8	12	.150	.240	1 9/32	1 5/32	5/8	1/4	.180	.181	1 5/32	9/32	5/8	1/4	.180	.181

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TABLE XI.—*Thread plug gaging members, taper lock design, range ¼ to 1½ inches, inclusive*



Not less than three full threads must remain on "not go" plug

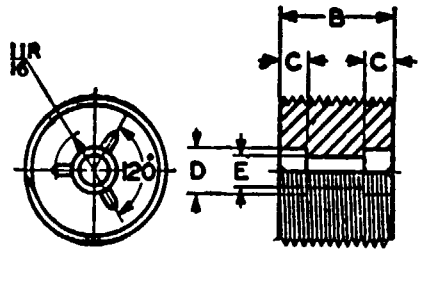
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Handle size No.	Range				Threads per inch	General dimensions										
	Thread plug diameters					Go						Not go				
	Nominal range, inclusive		Decimal range			A	B	C	D	E		A	B	C	D	E
	From—	To—	Above—	To and including—						Min.	Max.					Min.	Max.
	Ins.	Ins.	Ins.	Ins.		Inches	Inches	Inch	Inch	Inch	Inch	Inches	Inch	Inch	Inch	Inch	Inch
1	¼	5/16	0.240	0.365		1½	½	¾	¼	0.239	0.240	1 5/16	5/16	¾	¼	0.239	0.240
2	5/16	½	.365	.510		1¾	¾	¾	¼	.309	.310	1⅝	⅜	¾	¼	.309	.310
3	9/16	¾	.510	.825		1⅞	⅞	¾	¼	.408	.410	1½	½	¾	¼	.408	.410
4	⅞	1⅛	.825	1.135		2 3/16	1	⅞	5/16	.608	.610	1 13/16	⅝	⅞	5/16	.608	.610
5	1¼	1½	1.135	1.510	Coarser than 12	2⅝	1¼	1	⅜	.808	.810	2⅛	¾	1	⅜	.808	.810
5	1¼	1½	1.135	1.510	12 and finer	2⅜	1	1	⅜	.808	.810	2⅛	¾	1	⅜	.808	.810
5 [1,2]	1½	2	1.510	2.010	Coarser than 12	2⅞	1½	1	⅜	.808	.810	2⅛	¾	1	⅜	.808	.810
5 [1,2]	1½	2	1.510	2.010	12 and finer	2⅜	1	1	⅜	.808	.810	2⅛	¾	1	⅜	.808	.810
5 [2]	1½	2	1.510	2.010	7 and coarser	3¼	1⅞	1	⅜	.808	.810	2¼	⅞	1	⅜	.808	.810
5 [2]	1½	2	1.510	2.010	Finer than 7, coarser than 16.	2⅝	1¼	1	⅜	.808	.810	2¼	⅞	1	⅜	.808	.810
5 [2]	1½	2	1.510	2.010	16 and finer	2¼	⅞	1	⅜	.808	.810	2¼	⅞	1	⅜	.808	.810
5 [2]	2	2½	2.010	2.510	7 and coarser	3⅜	2	1	⅜	.808	.810	2¼	⅞	1	⅜	.808	.810
5 [2]	2	2½	2.010	2.510	Finer than 7, coarser than 16.	2¾	1⅜	1	⅜	.808	.810	2¼	⅞	1			
5 [2]	2	2½	2.010	2.510	16 and finer	2¼	⅞	1	⅜	.808	.810	2¼	⅞	1	⅜	.808	.810

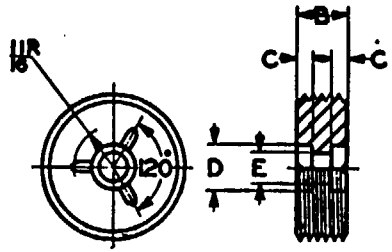
[1] These blanks are intended for the American National coarse and fine thread series only.
[2] The use of taper lock thread plug gaging members and handles is optional for straight thread plug gages in the ranges above 1.510 to and including 2.510 inches, but the reversible design is recommended as standard.

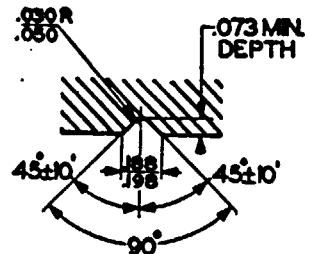
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TABLE XII.—*Thread plug gaging members, reversible design, range above 1½ to and including 2½ inches*

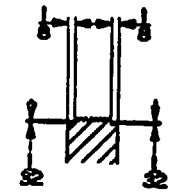


NOT LESS THAN THREE FULL THREADS MUST REMAIN ON "NOT GO" PLUG.





ENLARGED SECTION OF INTERNAL LOCKING MEMBER 3 REQUIRED ON BOTH ENDS OF ALL GAGES.

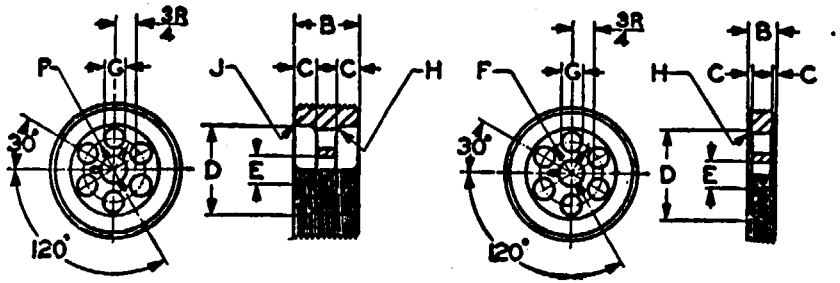


"NOT GO" GAGES 16 T.P.I. & FINER RELIEVED ON BOTH ENDS 32 BELOW SHARP ROOT OF THRD. AS SHOWN.

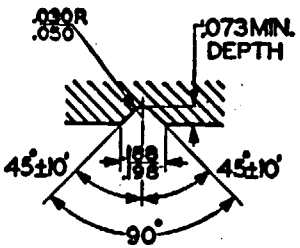
Handle size No.	Range, thread plug diameters		Go												Not go			
	Above—	To and including—	7 threads per inch and coarser				Finer than 7 threads per inch and coarser than 16				16 threads per inch and finer				All pitches			
			B	*C*	*D*	*E*	*B*	*C*	*D*	*E*	*B*	*C*	*D*	*E*	*B*	*C*	*D*	*E*
	Inches	*Inches*	*Inches*	*Inch*	*Inch*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch.*
6	1.510	2.010	1 7/8	1/2	25/32	17/32	1 1/4	3/8	25/32	17/32	7/8	9/32	25/32	17/32	7/8	9/32	25/32	17/32
6	2.010	2.510	2	1/2	25/32	17/32	1 5/8	3/8	25/32	17/32	7/8	9/32	25/32	17/32	7/8	9/32	25/32	17/32

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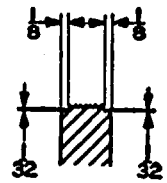
TABLE XIII.—*Thread plug gaging members, reversible design, range above 2½ to and including 4½ inches*



NOT LESS THAN THREE FULL THREADS MUST REMAIN ON "NOT GO" PLUG.



ENLARGED SECTION OF LOCKING GROOVE 3 REQUIRED ON BOTH ENDS OF ALL GAGES.

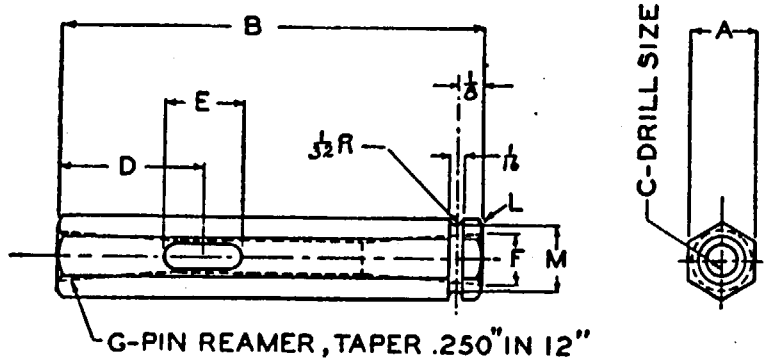


"NOT GO" GAGES 16 T.P.I. & FINER RELIEVED ON BOTH ENDS 32 BELOW SHARP ROOT OF THD AS SHOWN.

Handle size No.	Thread plug diameters				Go											Not go			All			
	Nominal range, inclusive		Decimal range		7 threads per inch and coarser				Finer than 7 threads per inch and coarser than 16				16 threads per inch and finer			All pitches			All pitches			
	From—	To—	Above—	To and including—	B	C	H	J	B	C	H	J	B	C	H	B	C	H	D	E	F	G
	Inches	*Ins.*	*Inches*	*Inches*	*Ins.*	*In.*	*In.*	*In.*	*Ins.*	*In.*	*In.*	*In.*	*In.*	*In.*	*In.*	*In.*	*In.*	*In.*	*Ins.*	*In.*	*Ins.*	*In.*
7	2½	3	2.510	3.010	2¼	1 1/16	3/16	1/16	1½	5/8	3/16	1/16	1	1/8	3/16	1	1/8	3/16	1⅞	2 9/32	-----	-----
7	3	3½	3.010	3.510	2¼	3/4	3/16	1/16	1½	5/8	1/4	1/16	1	1/8	3/16	1	1/8	3/16	2¼	2 9/32	-----	-----
7	3½	4	3.510	4.010	2¼	3/4	5/16	1/16	1½	5/8	1/4	1/16	1	1/8	3/16	1	1/8	3/16	2⅝	2 9/32	-----	-----
7	4	4½	4.010	4.510	2¼	3/4	5/16	1/16	1½	5/8	1/4	1/16	1	1/8	3/16	1	1/8	3/16	3	2 9/32	1 1/16	3/4

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TABLE XIV.—*Thread plug gage handles, taper lock design, range No. 0 to No. 12, inclusive*

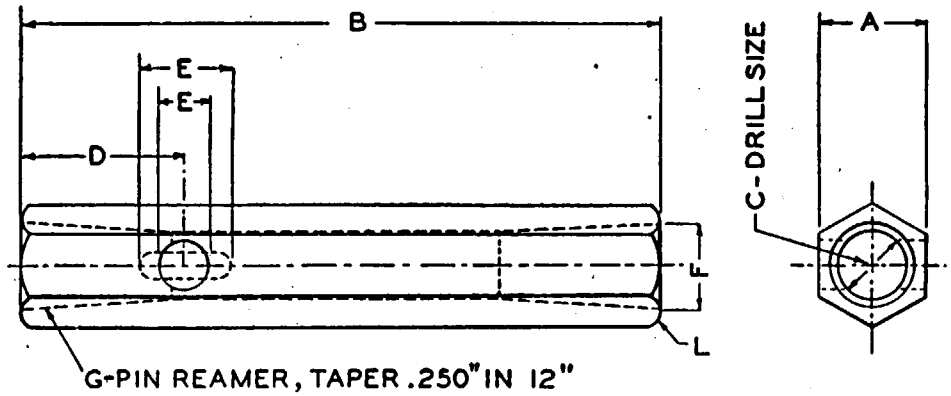


Handle size No.	Thread plug diameters				General dimensions										
	Nominal range, inclusive		Decimal range		*A*	*B*	*C* Drill size	*D*	*E*	F		*G*	*L*	*M*	
	From—	To—	Above—	To and including—						Min.	Max.			Min.	Max.
	No.	*No.*	*Inch*	*Inch*	*Inch*	*Inches*		*Inch*	*Inch*	*Inch*	*Inch*	*No.*	*Inch*	*Inch*	*Inch*
000	0	3	0.059	0.105	5/16	1 1/2	No. 34 (0.111)	9/16	5/64 by 1/4	0.125	0.126	000	1/32	0.172	0.177
00	4	6	.105	.150	1/4	1 3/4	No. 29 (0.136)	5/8	3/32 by 5/16	.155	.156	0	1/32	.235	.240
0	8	12	.150	.240	5/16	2	No. 20 (0.161)	11/16	1/8 by 3/8	.180	.181	2	1/32	.297	.302

NOTE.—The purpose of the groove in the "not go" end of the handle is to distinguish the "not go" from the "go" end.

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TABLE XV.—*Thread plug gage handles, taper lock design, range ¼ to 1½ inches, inclusive*

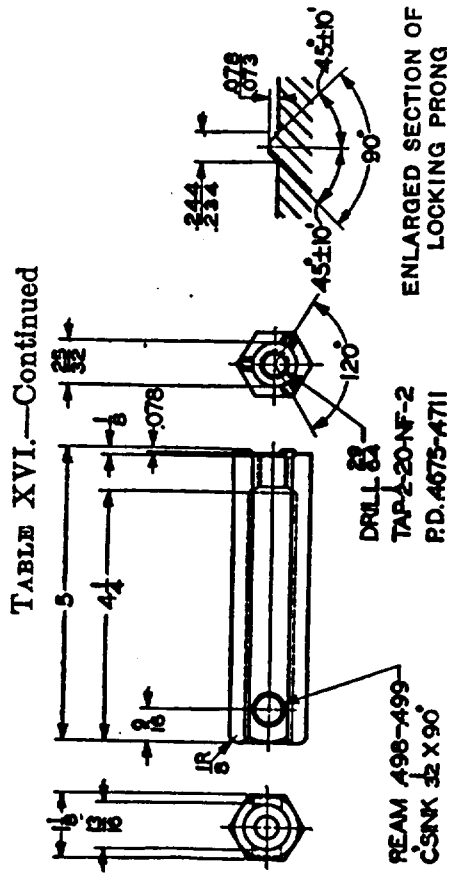


Handle size No.	Range (inclusive), thread plug diameters		General dimensions								
	From—	To—	A	B	C Drill size	D	E	F Min.	F Max.	G	L
	Inches	*Inches*	*Inches*	*Inches*		*Inches*	*Inch*	*Inch*	*Inch*	*No.*	*Inch*
1	¼	9/16	⅜	2¾	7/32	25/32	⅛ by ½	0.239	0.240	4	1/16
2	⅜	½	½	3	L (0.290)	25/32	15/64 diam.	.309	.310	6	1/16
3	9/16	¾	1 1/16	3¼	25/64	27/32	1 1/32 diam.	.409	.410	7	3/32
4	⅞	1⅛	⅞	3⅝	27/64	63/64	⅜ diam.	.609	.610	10	5/32
	1¼	1½	1⅛	4	25/32	1⅛	7/16 diam.	.809	.810	11	⅛
5	Above 1.510 (See note.)	2.510	1⅛	4	25/32	1⅛	7/16 diam.	.809	.810	11	⅛

NOTE.—In the range above 1.510 to and including 2.510 inches the No. 5 handle is optional. See footnote, Table XI.

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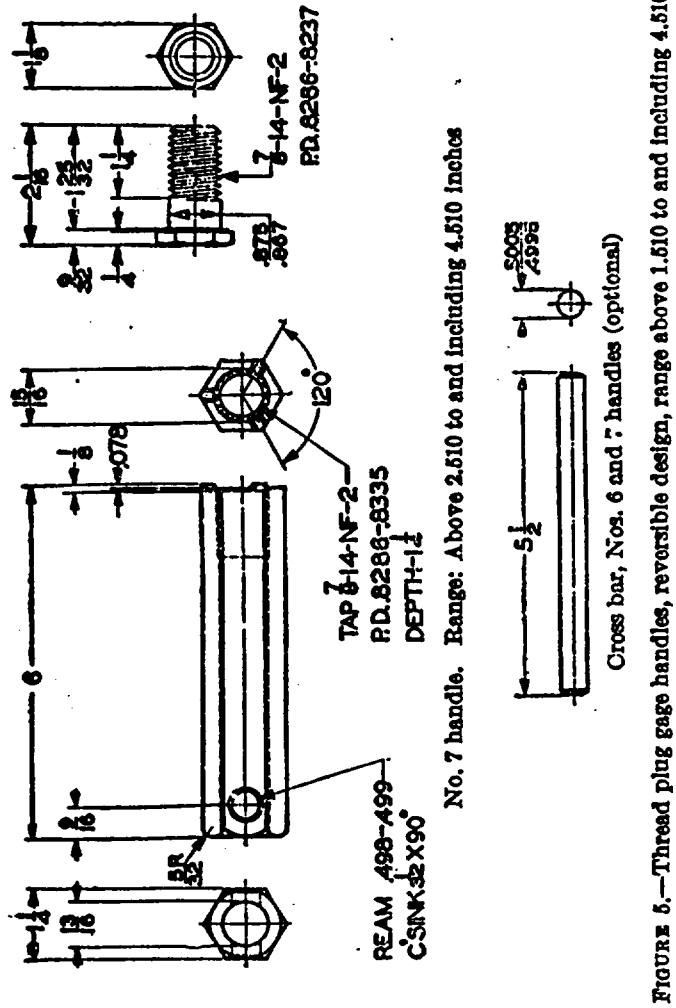
TABLE XVI.—Continued



Plug	B	C	E
	Inches	*Inch*	*Inches*
Not go, all pitches. Go, 16 threads per inch and finer	1½	5/16	1¼
Go, pitches coarser than 16 threads per inch	2¼	¼	1½

No. 6 handle. Range: Above 1.510 to and including 2.610 inches

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No. 7 handle. Range: Above 2.510 to and including 4.510 inches

Cross bar, Nos. 6 and 7 handles (optional)

FIGURE 5.—Thread plug gage handles, reversible design, range above 1.510 to and including 4.510 inches

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TABLE XVI.—*Tolerances for class X "go" thread gages and all "not go" thread gages*

Threads per inch	Tolerance on pitch diameter [1]		Tolerance in lead [2]	Tolerance on half angle of thread		Tolerance on major or minor diameters [1]	
	From—	To—				From—	To—
1	2	3	4	5		6	7
	Inch	*Inch*	*Inch*	°	′	*Inch*	*Inch*
80	0.0000	0.0002	±0.0002	±0	30	0.0000	0.0003
72	.0000	.0002	±.0002	±0	30	.0000	.0003
64	.0000	.0002	±.0002	±0	30	.0000	.0004
56	.0000	.0002	±.0002	±0	30	.0000	.0004
48	.0000	.0002	±.0002	±0	30	.0000	.0004
44	.0000	.0002	±.0002	±0	20	.0000	.0004
40	.0000	.0002	±.0002	±0	20	.0000	.0004
36	.0000	.0002	±.0002	±0	20	.0000	.0004
32	.0000	.0003	±.0003	±0	15	.0000	.0004
28	.0000	.0003	±.0003	±0	15	.0000	.0005
24	.0000	.0003	±.0003	±0	15	.0000	.0005
20	.0000	.0003	±.0003	±0	15	.0000	.0005
18	.0000	.0003	±.0003	±0	10	.0000	.0005
16	.0000	.0003	±.0003	±0	10	.0000	.0006
14	.0000	.0003	±.0003	±0	10	.0000	.0006
13	.0000	.0003	±.0003	±0	10	.0000	.0006
12	.0000	.0003	±.0003	±0	10	.0000	.0006
11	.0000	.0003	±.0003	±0	10	.0000	.0006
10	.0000	.0003	±.0003	±0	10	.0000	.0006
9	.0000	.0003	±.0003	±0	10	.0000	.0007
8	.0000	.0004	±.0004	±0	5	.0000	.0007
7	.0000	.0004	±.0004	±0	5	.0000	.0007
6	.0000	.0004	±.0004	±0	5	.0000	.0008
5	.0000	.0004	±.0004	±0	5	.0000	.0008
4½	.0000	.0004	±.0004	±0	5	.0000	.0008
4	.0000	.0004	±.0004	±0	5	.0000	.0009
3½	.0000	.0004	±.0004	±0	5	.0000	.0009

[1] On "go" plugs the tolerance is plus and on "go" rings the tolerance is minus. On "not go" plugs the tolerance is minus and on "not go" rings the tolerance is plus.
[2] Allowable variation in lead between any two threads not farther apart than the standard length of engagement, which is equal to the basic major diameter.

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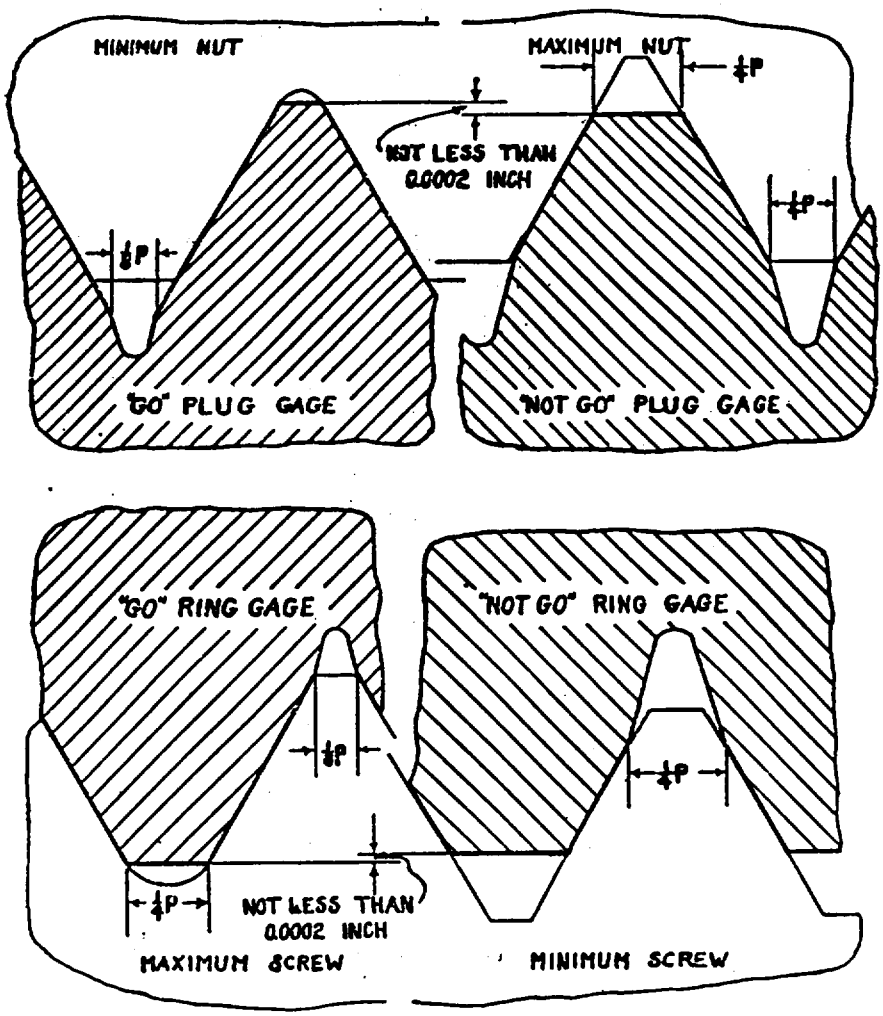
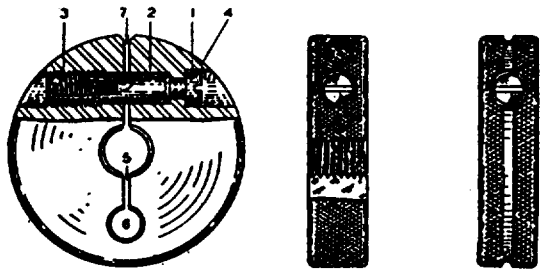
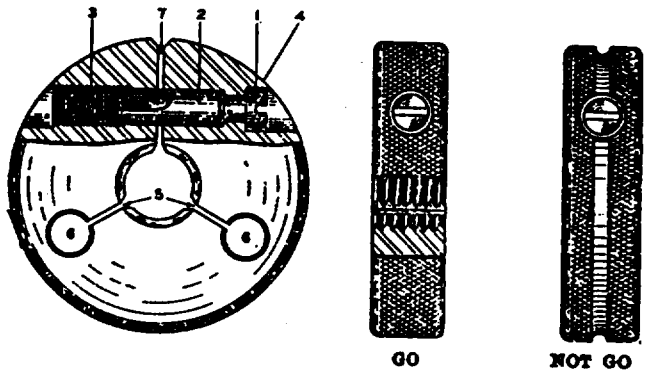


FIGURE 6.—Thread form of gages

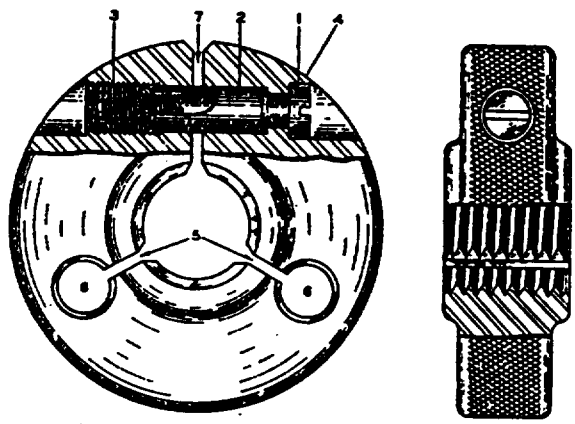
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:36Z
Check the source to verify that this is the current version before use.



Range: No. 0 to and including 0.365 inch, "go" and "not go" gages, all pitches



Range: 0.365 to and including 0.510 inch, "go" and "not go" gages, all pitches; 0.510 to and including 4.510 inches, "go" and "not go" gages, fine pitches; 0.510 to and including 4.510 inches, "not go" gages only, coarse pitches



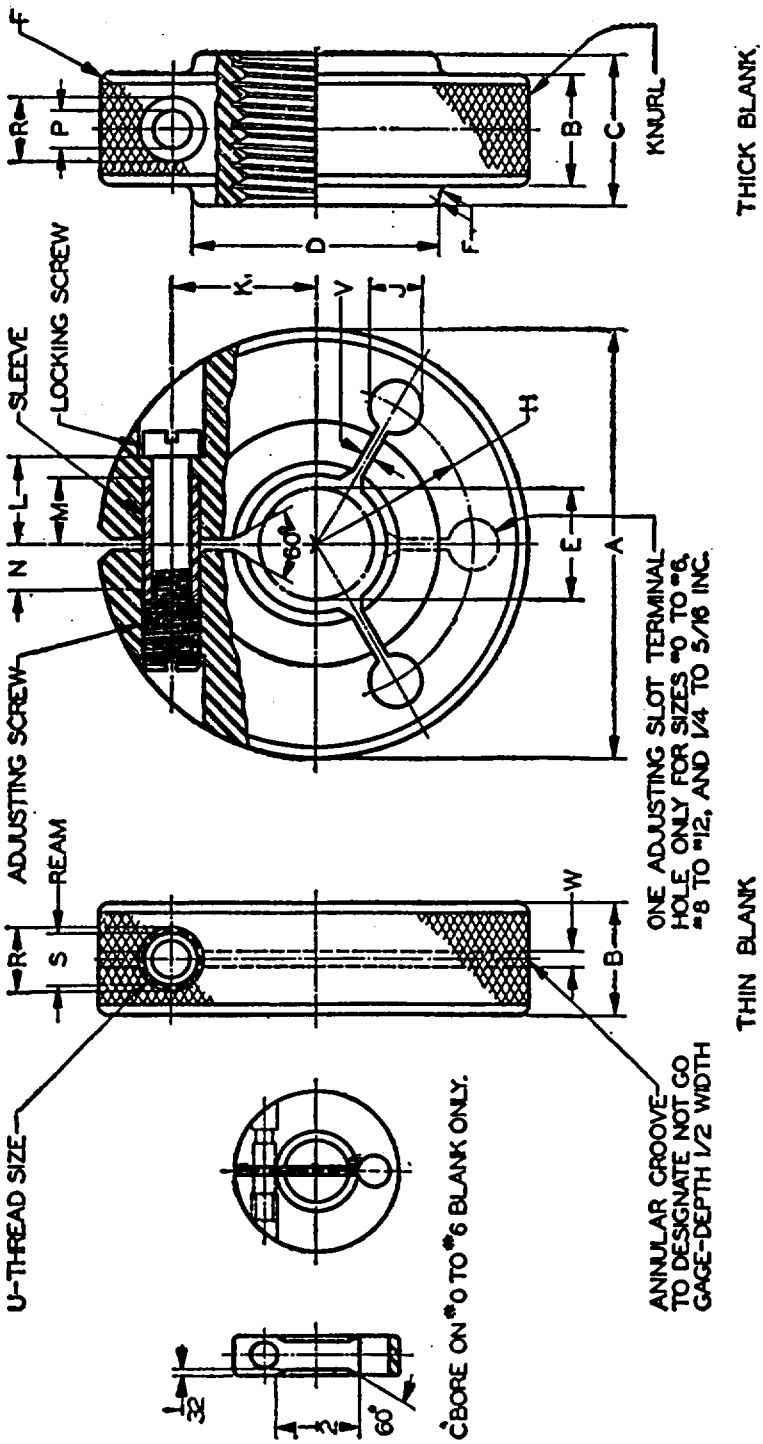
Range: 0.510 to and including 4.510 inches, "go" gages only, coarse pitches

FIGURE 7.—American gage design standard thread ring gages, details of construction, range No. 0 to 4½ inches, inclusive

1. Locking screw.
2. Sleeve.
3. Adjusting screw.
4. Body.
5. Adjusting slot.
6. Adjusting slot terminal hole.
7. Locking slot.

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TABLE XVII.—*Thread ring gages, range No. 0 to 4½ inches, inclusive*



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FEDERAL STANDARD STOCK CATALOGUE

(Section IV, Part 5)

Nominal range, inclusive	Decimal range, above and including—	General dimensions											
		A	*B*	*C*	*D*	*E*	*F*	*H*	*J*	*K*	*L*	*M*	*N*
	Ins.	*Ins.*	*In.*	*Ins.*	*Ins.*	*Ins.*	*In.*	*Ins.*	*In.*	*Ins.*	*In.*	*In.*	*In.*
Nos. 0 to 6	0.059 .150	1	1/4	------	------	------	1/32	5/16	5/32	5/16	7/32	5/32	1/16
Nos. 8 to 12	.150 .240	1	1/4	------	------	------	1/32	5/16	5/32	5/16	7/32	5/32	1/16
1/4 to 5/16	.240 .365	1 3/8	1 1/32	------	------	5/32	1/32	7/16	3/16	3/8	1 1/32	1/4	1/8
3/8 to 1/2	.365 .510	1 3/4	7/16	------	------	3/16	3/64	19/32	1/4	1 5/32	1/2	3/8	3/16
9/16 to 3/4	.510 .825	2 3/16	9/16	3/4	1 1/16	1 1/32	1/16	3/4	5/16	1 1/16	1 7/32	1 3/32	7/32
7/8 to 1 1/8	.825 1.135	2 5/8	1 1/16	1 5/16	1 1/2	9/16	1/16	3 1/32	5/16	7/8	1 7/32	1 3/32	9/32
1 1/4 to 1 1/2	1.135 1.510	3 1/4	3/4	1 1/8	1 7/8	2 7/32	1/16	1 3/16	3/8	1 1/8	3/4	7/16	5/16
1 5/8 to 2	1.510 2.010	3 3/4	1 3/16	1 1/4	2 3/8	1 3/16	3/32	1 7/16	3/8	1 3/8	3/4	7/16	5/16
2 1/8 to 2 1/2	2.010 2.510	4 1/2	7/8	1 5/16	2 7/8	1 19/32	3/32	1 3/4	7/16	1 11/16	1 5/16	9/16	7/16
2 5/8 to 3	2.510 3.010	5	7/8	1 3/8	3 3/8	2	3/32	2	7/16	1 15/16	1 5/16	9/16	7/16
3 1/8 to 3 1/2	3.010 3.510	5 1/2	1 5/16	1 7/16	3 7/8	2 7/16	3/32	2 7/32	7/16	2 3/16	1 5/16	9/16	7/16
3 5/8 to 4	3.510 4.010	6 3/8	1 5/16	1 1/2	4 3/8	2 15/16	3/32	2 5/8	1/2	2 9/16	1	3/4	5/16
4 1/8 to 4 1/2	4.010 4.510	7 1/4	1	1 1/2	5 3/8	3 3/8	3/32	3 1/32	1/2	3	1	3/4	5/16

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Nominal range, inclusive	General dimensions								
	P Drill size	*R* Drill size	*S* Ream		*U*			*V*	*W*
			Min.	Max.	Size	Pitch diameter			
						Min.	Max.		
			In.	*In.*		*In.*	*In.*	*In.*	*In.*
Nos. 0 to 6	No. 41 (0.0960)	11/64 (0.1719)	0.1370	0.1373	No. 8-36	0.1460	0.1478	0.010 (1)	1/32
Nos. 8 to 12	No. 41 (0.0960)	11/64 (0.1719)	.1370	.1373	No. 8-36	.1460	.1478	1/64	1/32
1/4 to 5/16	No. 31 (0.1200)	7/32 (0.2187)	.1810	.1813	No. 12-28	.1928	.1950	1/32	1/16
5/16 to 1/2	No. 25 (0.1495)	17/64 (0.2656)	.2150	.2153	1/4-28	.2268	.2290	1/32	3/32
9/16 to 3/4	No. 7 (0.2010)	21/64 (0.3281)	.2720	.2723	5/16-24	.2854	.2878	1/16	3/32
7/8 to 1 1/16	No. 1 (0.2280)	35/64 (0.3906)	.3340	.3344	3/8-24	.3479	.3503	1/16	3/32
1 1/4 to 1 1/2	17/64 (0.2656)	29/64 (0.4531)	.3890	.3894	7/16-20	.4050	.4076	1/16	3/32
1 5/8 to 2	17/64 (0.2656)	29/64 (0.4531)	.3890	.3894	7/16-20	.4050	.4076	1/16	1/8
2 1/8 to 2 1/2	21/64 (0.3281)	33/64 (0.5156)	.4510	.4515	1/2-20	.4675	.4701	3/32	1/8
2 5/8 to 3	21/64 (0.3281)	33/64 (0.5156)	.4510	.4515	1/2-20	.4675	.4701	3/32	1/8
3 1/8 to 3 1/2	21/64 (0.3281)	33/64 (0.5156)	.4510	.4515	1/2-20	.4675	.4701	3/32	1/8
3 5/8 to 4	13/32 (0.4062)	41/64 (0.6406)	.5710	.5715	5/8-18	.5889	.5919	3/32	1/8
4 1/8 to 4 1/2	13/32 (0.4062)	41/64 (0.6406)	.5710	.5715	5/8-18	.5889	.5919	3/32	1/8

[1] Approximate.

NOTE.—Thin gage blanks are to be used for all "not go" thread ring gages, all American National fine-thread "go" ring gages, and American National coarse-thread "go" ring gages up to and including the 1/2-inch size. Thick blanks are to be used for all American National coarse-thread "go" ring gages above the 1/2-inch size.

It is recommended that standard thread ring gage blanks be used for special pitches as follows:

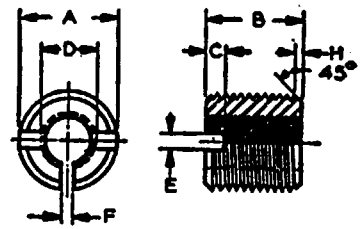
No. 0 to 1/2 inch, inclusive, thin blanks for all pitches.

Above 1/2 inch to and including 1 1/16 inches, thin blanks for 12 threads per inch and finer, and thick blanks for all "go" gages in pitches coarser than 12 threads per inch.

Above 1 1/16 inches, thin blanks for 10 threads per inch and finer, and thick blanks for all "go" gages in pitches coarser than 10 threads per inch.

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TABLE XVIII.—*Thread ring gage adjusting screws*

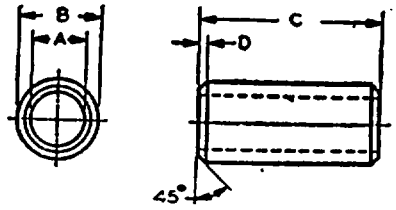


A			Minor diameter		B[1]	C	D				E	F	H
Size	Pitch diameter		Minimum	Maximum			Size	Pitch diameter		Tap drill			
	Minimum	Maximum						Minimum	Maximum				
	Inch	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*		*Inch*	*Inch*	*No.*	*Inch*	*Inch*	*Inch*
No. 8–36	0.1442	0.1460	0.1315	0.1333	3/16	3/64	No. 2–64	0.0759	0.0773	50 (0.070)	1/32	1/64	0.020
No. 12–28	.1906	.1928	.1744	.1766	1/4	3/64	No. 4–48	.0985	.1001	42 (.093)	1/32	1/64	.020
1/4–28	.2246	.2268	.2084	.2106	5/16	1/16	No. 6–40	.1218	.1235	32 (.116)	3/64	1/32	.020
5/16–24	.2830	.2854	.2641	.2665	5/16	1/16	No. 10–32	.1697	.1716	20 (.161)	3/64	1/32	1/32
3/8–24	.3455	.3479	.3266	.3290	3/8	3/64	No. 12–28	.1928	.1950	14 (.182)	1/16	3/64	1/32
7/16–20	.4024	.4050	.3797	.3823	7/16	3/64	1/4–28	.2268	.2290	3 (.213)	1/16	3/64	1/32
1/2–20	.4649	.4675	.4422	.4448	1/2	3/32	5/16–24	.2854	.2878	I (.272)	3/64	1/16	3/64
5/8–18	.5859	.5889	.5607	.5637	5/16	3/32	3/8–24	.3479	.3503	Q (.332)	3/64	1/16	3/64

[1] Tolerance on length $B = \pm 1/64$ inch.

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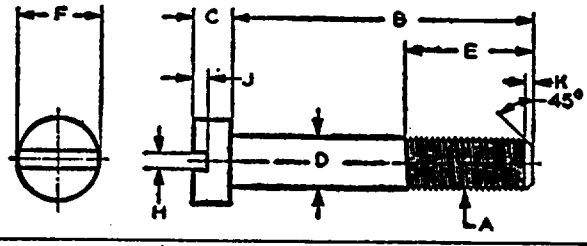
TABLE XIX.—*Thread ring gage sleeves*



A	B		C[1]	D
	Minimum	Maximum		
	Inch	*Inch*	*Inches*	*Inch*
No. 43 (0.089)	0.1368	0.1370	1/4	0.010
No. 32 (.116)	.1808	.1810	5/16	.020
No. 27 (.144)	.2148	.2150	5/8	.020
No. 10 (.193)	.2718	.2720	13/16	1/32
No. 2 (.221)	.3337	.3340	3/4	1/32
F (.257)	.3887	.3890	13/16	1/32
P (.323)	.4507	.4510	1 1/16	3/64
25/64 (.391)	.5707	.5710	1 1/2	3/64

[1] Tolerance on length C = ±1/64 inch.

TABLE XX.—*Thread ring gage locking screws*



A			B[1]	C	D		E	F	H	J	K
Size	Pitch diameter										
	Min.	Max.			Min.	Max.					
	Inch	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*
No. 2-64	0.0745	0.0759	29/64	5/64	0.0840	0.0860	3/16	5/32	1/32	3/64	0.010
No. 4-48	.0969	.0985	23/32	3/32	.1100	.1120	5/16	5/16	1/32	3/64	.020
No. 6-40	.1201	.1218	1	1/8	.1360	.1380	7/16	7/32	3/64	1/16	.020
No. 10-32	.1678	.1697	1 1/16	1/8	.1880	.1900	7/16	9/32	3/64	1/16	1/32
No. 12-28	.1906	.1928	1 3/16	5/32	.2140	.2160	1/2	11/32	1/16	1/16	1/32
1/4-28	.2246	.2268	1 23/64	5/16	.2480	.2500	9/16	13/32	1/16	3/64	1/32
5/16-24	.2830	.2854	1 23/32	1/4	.3105	.3125	5/8	15/32	3/64	3/32	3/64
3/8-24	.3455	.3479	2 3/16	5/16	.3730	.3750	3/4	19/32	3/64	3/32	3/64

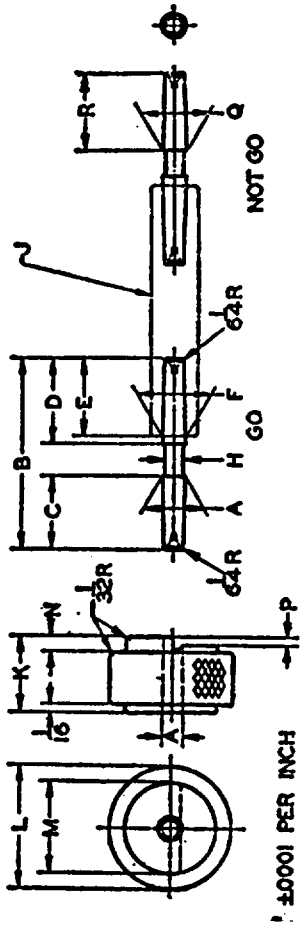
[1] Tolerance on length B = −1/32 inch.

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FEDERAL STANDARD STOCK CATALOGUE

(Section IV, Part 5)

ecking handles and gaging members of taper lock plug gages, range .059 to and including 0.240 inch



AGES FOR CHECKING HANDLES

Go and not go									Handle No.
R ±0.0002	B	D	E	F		H	J		
				Min.	Max.				
Inch	Inches	Inch	Inch	Inch		Inch			
0.8000	1¾	9/16	½	0.125	0.128	3/32	000		
.5625	1 17/32	⅝	5/16	.155	.156	¼	00		
.6250	1 17/32	1 1/16	⅝	.190	.181	9/64	0		

FEDERAL STANDARD STOCK CATALOGUE

(Section IV, Part 5)

ages, range

dle	Q
	Inch
1	0.025
2	.025
3	.050
4	.100
5	.200

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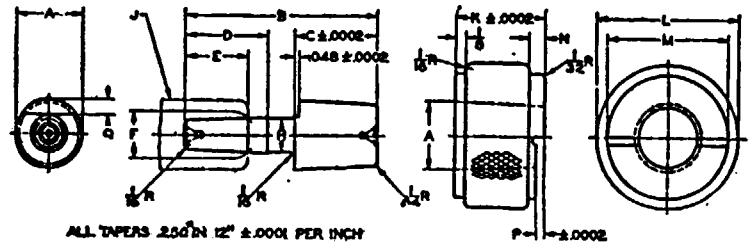
RING GAGES FOR CHECKING GAGING MEMBERS

Size	A	K	L	M	N	P
		±0.0002				±0.0002
	Inch	Inch	Inches	Inch	Inch	Inch
000	0.126	0.5480	1	3/4	1/8	0.0480
00	.156	.6105	1	3/4	1/8	.0480
0	.181	.6730	1 1/8	7/8	1/8	.0480

NOTE.—In the range above 0.059 to and including 0.240 inch, the taper limits established by the American Gage Design Committee for handles and shanks may be readily maintained by the use of the gages shown above, in which the taper-plug gage is a double-end limit gage. Equally satisfactory results may be secured using a single-end taper-plug gage in which a scribed line represents the minimum size of hole and the shoulder of the gage represents the maximum size of hole.

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TABLE XXII.—*Plug and ring gages for checking handles and gaging members of taper lock plug gages, range above 0.240 to and including 1.510 inches*



PLUG GAGES FOR CHECKING HANDLES

Handle size No.	A	B	C	D	E	F		H	Handle No.	Q
						Min.	Max.		J	
	Inch	Inches	Inch	Inch	Inch	Inch	Inch	Inch		Inch
1	0.240	2	0.7500	1	3/4	0.239	0.240	3/16	1	0.025
2	.310	2	.7500	1	3/4	.239	.240	1/4	2	.025
3	.410	2	.7500	1	3/4	.309	.310	5/16	3	.050
4	.610	2 3/16	.8750	1	3/4	.408	.410	13/32	4	.100
5	.810	2 3/16	1.0000	1	3/4	.408	.410	13/32	5	.200

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RING GAGES FOR CHECKING GAGING MEMBERS

Size	A	K	L	M	N	P
	Inch	Inches	Inches	Inches	Inch	Inch
1	0.240	0.7980	1⅛	⅞	5/32	0.0480
2	.310	.7980	1¼	1	5/32	.0480
3	.410	.8460	1⅜	1⅛	3/16	.0960
4	.610	.9710	1 9/16	1 5/16	3/16	.0960
5	.810	1.0960	1¾	1½	3/16	.0960

NOTE.—In the range above 0.240 to and including 1.510 inches, the taper limits established by the American Gage Design Committee for taper lock handles and shanks may be readily maintained by the use of the gages shown above, in which the taper plug gage is of the single-end limit type, with a ground step representing the minimum size hole. Equally satisfactory results may be secured using a single-end taper-plug gage, in which a scribed line represents the minimum size of hole and the shoulder of the gage represents the maximum size of hole.

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TABLE XXIII.—*Tolerances for class Y "go" thread gages*

Threads per inch	Tolerance on pitch diameter [1]		Tolerance in lead [2]	Tolerance on half angle of thread		Tolerance on major or minor diameters [1]	
	From—	To—				From—	To—
1	2	3	4	5		6	7
	Inch	*Inch*	*Inch*	°	′	*Inch*	*Inch*
80	0.0001	0.0003	±0.0002	±0	45	0.0000	0.0003
72	.0001	.0003	±.0002	±0	45	.0000	.0003
64	.0001	.0004	±.0002	±0	45	.0000	.0004
56	.0001	.0004	±.0002	±0	45	.0000	.0004
48	.0001	.0004	±.0002	±0	45	.0000	.0004
44	.0001	.0004	±.0002	±0	30	.0000	.0004
40	.0001	.0004	±.0002	±0	30	.0000	.0004
36	.0001	.0004	±.0002	±0	30	.0000	.0004
32	.0001	.0004	±.0003	±0	20	.0000	.0004
28	.0002	.0005	±.0003	±0	20	.0000	.0005
24	.0002	.0005	±.0003	±0	20	.0000	.0005
20	.0002	.0005	±.0003	±0	20	.0000	.0005
18	.0002	.0005	±.0003	±0	15	.0000	.0005
16	.0002	.0006	±.0003	±0	15	.0000	.0006
14	.0002	.0006	±.0003	±0	15	.0000	.0006
13	.0002	.0006	±.0003	±0	15	.0000	.0006
12	.0002	.0006	±.0003	±0	10	.0000	.0006
11	.0002	.0006	±.0003	±0	10	.0000	.0006
10	.0002	.0006	±.0003	±0	10	.0000	.0006
9	.0002	.0007	±.0003	±0	10	.0000	.0007
8	.0002	.0007	±.0004	±0	5	.0000	.0007
7	.0002	.0007	±.0004	±0	5	.0000	.0007
6	.0003	.0008	±.0004	±0	5	.0000	.0008
5	.0003	.0008	±.0004	±0	5	.0000	.0008
4½	.0003	.0008	±.0004	±0	5	.0000	.0008
4	.0003	.0009	±.0004	±0	5	.0000	.0009
3½	.0003	.0009	±.0004	±0	5	.0000	.0009

[1] On "go" plugs the tolerance is plus and on "go" rings the tolerance is minus.
[2] Allowable variation in lead between any two threads not farther apart than the standard length of engagement, which is equal to the basic major diameter.

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TABLE XXIV.—*Tolerances for class Z "go" thread gages*

Threads per inch	Tolerance on pitch diameter [1]		Tolerance in lead [2]	Tolerance on half angle of thread	Tolerance on major or minor diameters [1]	
	From—	To—			From—	To—
1	2	3	4	5	6	7
	Inch	*Inch*	*Inch*	° ′	*Inch*	*Inch*
80	0.0002	0.0006	±0.0002	±0 45	0.0000	0.0003
72	.0002	.0006	±.0002	±0 45	.0000	.0003
64	.0002	.0006	±.0002	±0 45	.0000	.0004
56	.0002	.0007	±.0002	±0 45	.0000	.0004
48	.0002	.0007	±.0002	±0 45	.0000	.0004
44	.0002	.0007	±.0002	±0 30	.0000	.0004
40	.0002	.0007	±.0002	±0 30	.0000	.0004
36	.0003	.0008	±.0002	±0 30	.0000	.0004
32	.0003	.0008	±.0003	±0 20	.0000	.0004
28	.0003	.0008	±.0003	±0 20	.0000	.0005
24	.0003	.0009	±.0003	±0 20	.0000	.0005
20	.0003	.0009	±.0003	±0 20	.0000	.0005
18	.0004	.0010	±.0004	±0 15	.0000	.0005
16	.0004	.0010	±.0004	±0 15	.0000	.0006
14	.0004	.0010	±.0004	±0 15	.0000	.0006
13	.0004	.0011	±.0004	±0 15	.0000	.0006
12	.0004	.0011	±.0004	±0 10	.0000	.0006
11	.0004	.0011	±.0004	±0 10	.0000	.0006
10	.0005	.0012	±.0004	±0 10	.0000	.0006
9	.0005	.0012	±.0004	±0 10	.0000	.0007
8	.0006	.0013	±.0005	±0 5	.0000	.0007
7	.0006	.0013	±.0005	±0 5	.0000	.0007
6	.0006	.0014	±.0005	±0 5	.0000	.0008
5	.0007	.0015	±.0005	±0 5	.0000	.0008
4½	.0007	.0015	±.0005	±0 5	.0000	.0008
4	.0007	.0016	±.0005	±0 5	.0000	.0009
3½	.0007	.0016	±.0005	±0 5	.0000	.0009

[1] On "go" plugs the tolerance is plus, and on "go" rings the tolerance is minus.
[2] Allowable variation in lead between any two threads not farther apart than the standard length of engagement, which is equal to the basic major diameter.

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